Hochschule Fulda
University of Applied Sciences

Global Distributed Software Development/
Teamprojekt (AI5088)

"Master Team Project Fall 2024"

Way2Home

Group 5 (Local)

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Milestone 2

(Dated: 11/21/2024)

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Revision History

Date Submitted	Date Revised	Revision Summary	Version
21/11/2024	29/11/2024	Changed Priorities of Features in “Functional Requirements”,Updated “Wireframe Images” and made minor adjustments	2.0

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1. **Executive Summary (Way2Home)**

Our team is developing **Way2Home**, a secure and user-focused web platform specifically designed to address the housing needs of Fulda University students. Recognizing the challenges students—particularly international ones—face in navigating the complex rental market, *Way2Home* provides a trusted and moderated environment tailored to their unique requirements. The platform offers advanced filtering options, allowing users to refine searches by price, location, pet-friendliness, accessibility, and neighborhood traits, ensuring students can quickly find housing that meets their specific needs. In addition, personalized notification alerts actively inform users of new listings that match their criteria, making the housing search efficient and stress-free.

What truly sets *Way2Home* apart is the deep understanding of student challenges brought by our team, which includes Fulda University students with firsthand experience navigating these issues. Designed to be accessible, media-rich, and responsive across devices, *Way2Home* delivers a seamless, engaging, and efficient user experience that exceeds the expectations of modern renters.

Key Advantages, Novelty, and Value

Way2Home distinguishes itself through its focus on verified listings, advanced filters, and user safety, setting it apart from other rental platforms. The **Tenant Score** ensures a tailored and efficient matchmaking process for landlords, while the **Personalized Notification Alerts** keep students updated on the newest and most relevant listings, reducing the time spent searching. These unique features, paired with our team's technical expertise and firsthand knowledge of student needs, allow us to provide a high-quality, secure rental experience, especially for those unable to visit properties in person.

Why Fund Way2Home

Supporting **Way2Home** means investing in a vital solution to Fulda University's housing challenges. With funding, we can enhance security, expand features, and scale our reach, ensuring more students benefit from this innovative platform. This investment not only simplifies housing for students but also creates a secure and trusted rental environment, addressing a critical need at a pivotal time.

2. Personas and Main Use Cases

Personas

Persona 1: Emily Li – The Student User (Primary Persona)

- **Characteristics**: Emily is a 20-year-old undergraduate international student, new to the area, looking for affordable, conveniently located housing close to campus. Balancing full-time studies, she values a streamlined experience that allows her to quickly find safe, reputable housing options.
- **Goals**: Quickly locate budget-friendly housing close to campus or transit; easily shortlist and contact landlords; save potential rentals to review later.
- **Skills**: Proficient with mobile and web platforms, accustomed to using filters and search functions, and comfortable with direct communication through in-app messaging.
- **Pain Points**: Time constraints due to academic demands; budget limitations; unreliable or misleading listings; occasional communication delays with landlords.

Persona 2: Michael Reyes – The Guest User

- **Characteristics**: Michael is a parent visiting from out of town, helping his daughter explore potential housing options. He's unfamiliar with the local rental market but wants an overview of availability and pricing.
- **Goals**: Browse current listings, gain an understanding of pricing, property types, and amenities; assist his daughter in finding suitable options.
- **Skills**: Basic web navigation, interested primarily in browsing without the need for interactive features.
- **Pain Points**: Limited access to property details and contact information as a guest; may need more functionality to explore listings in-depth, which could reduce initial engagement.

Persona 3: Linda Weber – The Landlord

- **Characteristics**: Linda owns a small property management business, renting out several apartments near the university. She values efficiency and relies on digital platforms to reach student tenants.
- **Goals**: Quickly create and manage listings, attract responsible tenants, and facilitate prompt, secure communication with interested students.

- **Skills**: Familiar with online platforms, comfortable managing listings, uploading images, and responding to inquiries, though not deeply technical.
- **Pain Points**: Time-consuming listing creation; challenges in ensuring listings are appealing to students; managing multiple listings and tenant communication effectively.

Persona 4: Alex Jansen – The Admin

- **Characteristics**: Alex is a Fulda University administrator with oversight of the platform, responsible for ensuring quality listings and compliance with university standards.
- **Goals**: Manage website functionality, oversee listing quality, ensure smooth operation and security.
- **Skills**: Skilled in platform administration, data handling, user management, and troubleshooting.
- **Pain Points**: Balancing content moderation with user satisfaction, overseeing a large volume of listings, maintaining security and platform integrity.

Persona 5: Julia Braun – The Moderator

- **Characteristics**: Julia works part-time as a moderator, tasked with reviewing content and ensuring listing quality. With experience in the local rental market, she understands students' needs well.
- **Goals**: Approve or reject listings, maintain listing quality, address user-reported issues or complaints.
- **Skills**: Familiar with web content management, detail-oriented, experienced in review and approval processes.
- **Pain Points**: Ensuring high standards for listings, handling flagged or inappropriate content, maintaining consistency in approvals.

Use Cases

UC-1: Guest Views Property Listings

- **Actor:** Michael Reyes (Guest)
- **Goal:** Explore basic housing information without an account.

- **Description:** Michael, as a guest, accesses the listings page and views basic property details like rent, location, and apartment photos. Since he doesn't have an account, he cannot view comple proper contact landlords or save listings. He can browse freely but has limited interaction with the platform's

features, and will need to create an account to access additional services like messaging or saving favorites.

UC-2: Student Explores and Searches Listings

- **Actor:** Emily Li (Student)
- **Goal:** Discover and explore housing options through browsing and targeted searches.
- **Description:**
  Emily logs into her account and navigates to the listings page to explore available properties. Initially, she scrolls through the unfiltered listings, viewing property details and photos to gain a broad sense of available options, including different areas and rental ranges.

  When Emily decides to refine her exploration, she uses the search fields to locate properties based on specific criteria, such as city names, neighborhoods, or amenities like "balcony" or "near subway." As she types, the application dynamically displays matching results, allowing her to quickly access relevant listings without scrolling through irrelevant options. This combination of browsing and targeted searching provides Emily with flexibility in discovering suitable properties.

UC-3: Student Filters Property Listings

- **Actor:** Emily Li (Student)
- **Goal:** Narrow down property search based on specific criteria.
- **Description:** Emily logs in, applies filters for location, rent, and furnished options, and reviews the listings that match her criteria. She can further refine her search based on availability and specific apartment features, making it easier to find a suitable place to rent. Emily also has the option to sort results by price, distance, or other relevant factors.

UC-4: Student Adds Listing to Favorites

- **Actor:** Emily Li (Student)
- **Goal:** Save listings for easy comparison and later access.
- **Description:** Emily saves specific listings to her “Favorites” list, allowing her to review and compare them later without reapplying her search filters. She

can access the favorites list from her profile, making it easy to revisit options before making a final decision.

UC-5: Student Views Nearby Amenities

- **Actor:** Emily Li (Student)
- **Goal:** Assess neighborhood suitability based on nearby amenities.
- **Description:** Emily opens the "Nearby Amenities" feature for a listing to check the proximity to cafes, transport hubs, and other points of interest. The feature includes a map view with distances to essential services, helping Emily make an informed decision about the property and its location.

UC-6: Landlord Creates a New Listing

- **Actor:** Linda Weber (Landlord)
- **Goal:** Post a new property for rent.
- **Description:** Linda logs in, fills out listing details, uploads photos, and submits the listing for moderator review. She can also add special instructions or features, such as pet policies or parking availability, to attract the right tenants. Once submitted, the listing will be reviewed and published after approval.

UC-7: Moderator Reviews and Approves Listing

- **Actor:** Julia Braun (Moderator)
- **Goal:** Ensure listing quality and compliance.
- **Description:** Julia reviews Linda's submitted listing, checks for clarity and quality, and provides feedback if needed. Once updated, Julia approves the listing, making it visible on the platform. She also ensures that the listing complies with the platform's policies and is free from misleading or inappropriate content.

UC-8: Student Contacts Landlord via In-App Messaging

- **Actor:** Emily Li (Student)
- **Goal:** Inquire about a specific property.

- **Description:** Emily uses the secure in-app messaging system to ask Linda questions about the property, such as lease length and pet policy. The system allows for easy and direct communication between students and landlords without the need for third-party email or phone contact.

UC-9: Landlord Manages Listings and Responds to Inquiries

- **Actor:** Linda Weber (Landlord)
- **Goal:** Respond to potential tenant inquiries.
- **Description:** Linda checks her dashboard for messages on her active listings and responds to student inquiries, keeping all communications in one place. She can also update or modify her listings based on feedback or market changes and ensure timely responses to prospective tenants.

UC-10: Student Sets Housing Alert Notifications

- **Actor:** Emily Li (Student)
- **Goal:** Receive alerts for new listings matching criteria.
- **Description:** Emily enables “Housing Alerts” to notify her when new listings within her budget and preferred location are posted. She can adjust her notification preferences, ensuring she only receives alerts for properties that meet her specific search criteria.

UC-11: Moderator Addresses a User Complaint

- **Actor:** Julia Braun (Moderator)
- **Goal:** Investigate and resolve a user-reported issue.
- **Description:** Julia reviews a complaint from a student about a misleading listing and takes appropriate action after verifying details with the landlord. She ensures the issue is addressed by contacting the involved parties and resolving the matter in accordance with platform policies.

UC-12: Admin Monitors Platform Activity and Security

- **Actor:** Alex Jansen (Admin)
- **Goal:** Ensure platform integrity and security.

- **Description:** Alex checks site analytics, reviews flagged content, and addresses security concerns to maintain compliance and trustworthiness. He also manages user access, ensuring only authorized users can make changes, and handles any system issues that may arise.

3. List of Non-Functional Requirements

1. Application shall be developed, tested and deployed using tools and servers approved by Class CTO and as agreed in Milestone 0. Application delivery shall be from chosen cloud server
2. Application shall be optimized for standard desktop/laptop browsers e.g. must render correctly on the two latest versions of two major browsers
3. All or selected application functions must render well on mobile devices
4. Data shall be stored in the database on the team's deployment cloud server.
5. Full resolution free media shall be downloadable directly, and full resolution media for selling shall be obtained after contacting the seller/owner
6. No more than 50 concurrent users shall be accessing the application at any time
7. Privacy of users shall be protected, and all privacy policies will be appropriately communicated to the users.
8. The language used shall be English (no localization needed)
9. Application shall be very easy to use and intuitive
10. Application should follow established architecture patterns
11. Application code and its repository shall be easy to inspect and maintain
12. Google analytics shall be used (optional for Fulda teams)
13. No email clients shall be allowed.
14. Pay functionality, if any (e.g. paying for goods and services) shall not be implemented nor simulated in UI.
15. Site security: basic best practices shall be applied (as covered in the class) for main data items
16. Application shall be media rich (images, video etc.). Media formats shall be standard as used in the market today
17. Modern SE processes and practices shall be used as specified in the class, including collaborative and continuous SW development
18. For code development and management, as well as documentation like formal milestones required in the class, each team shall use their own GitHub to be set-up by class instructors and started by each team during Milestone 0
19. The application UI (WWW and mobile) shall prominently display the following exact text on all pages "Fulda University of Applied Sciences Software Engineering Project, Fall 2024 For Demonstration Only" at the top of the WWW page. (Important to not confuse this with a real application).

4. Competitive Analysis

Features	WG	kleinanzeigen	Immo Scout24	Way2Home
Guest User	Yes	Yes	Yes	Yes
Browse Listings	Yes	Yes	Yes	Yes
Basic Search	Yes	Yes	Yes	Yes
Advanced Search Filters	Yes	Yes	Yes	Yes
Verified Listings	No	No	Yes	Yes
Favoriting Listing	Yes	Yes	Yes	Yes
In-App Messaging	Yes	Yes	Yes	Yes
Personalized Alerts	No	No	Yes	Yes
Tenant Score	No	No	No	Yes
Video Chat	No	No	No	Yes

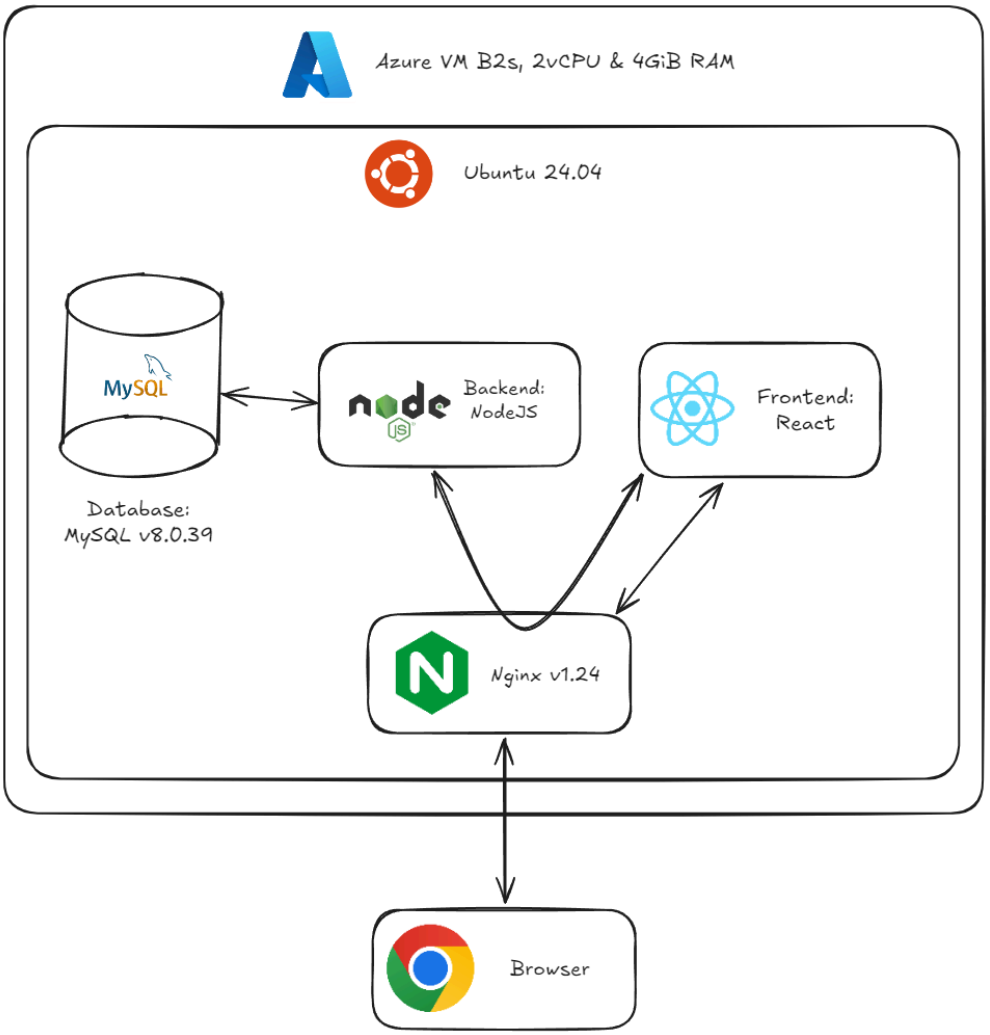
Communication				
Reviews & Ratings	No	No	Yes	Yes
Schufa Report	Yes	No	Yes	No
Real-Estate Support	No	No	Yes	No
Premium Membership	Yes	No	Yes	No
Multilingual Search	No	No	Yes	No

Way2Home offers essential features like browsing listings, saving favorites,messaging, and search filters to make the rental process smooth and easy. Unlike other platforms, Way2Home includes both verified listings and tenant scoring, giving landlords a better idea of tenant reliability. It also offers video chat, which is especially useful for international students to connect directly with landlords. Reviews and ratings further add transparency and trust to the rental process. With these unique features, Way2Home provides a student-friendly and secure platform for finding accommodation in Germany.

5. High-Level System Architecture and Technologies Used

Technologies

- Server host: Azure VM B2s, 2vCPUs 4 GiB RAM
- OS: Ubuntu 24.04
- Database: MySQL v8.0.39
- Web server: Nginx v1.24.0
- Server side language: Node.js
- Web framework: React
- IDE: Visual Studio Code
- SSL Cert: Let's Encrypt (certbot v2.9.0)



Deployment Information

Supported browsers are Chrome 130 & Firefox 132. Deployment is done via GitHub actions. The action builds the software, moves it to the server via SCP, and then restarts the service to serve the new version.

6. Team and Roles

Name	Role	Responsibilities
Romina Nikasa	Team Lead	Keep the team focussed on the goals, communication with Professor
Rashid Hussain Mohammed	GitHub Master	Organizes Git and makes sure commit policies are followed
Tim Schröder	Backend Lead	Review backend PRs
Anmoldeep Angadi	Frontend Lead	Review frontend PRs
Samarth Mahajan	Frontend Developer	Frontend side Development
Suhaila Kondappilly Aliyar	Backend Developer	Backend side Development
Maira Shafi	Documentation Master	Maintain Documentation, and overall Accuracy and Relevance

7. Checklist

Review Tasks: Answer the following items with either 'DONE,' 'ON TRACK,' or 'ISSUE':

Meeting schedule established	DONE
GitHub master chosen	DONE
Agreement on software tools	DONE
Readiness to use front-end/back-end frameworks	DONE
Team understanding of the final M1 document	DONE
Organization of the GitHub repository	DONE

8. Functional Requirements - Prioritized:

Priority 1 Requirements

Unregistered Users (Guests)

RequirementID: 4

Title: Basic Search Functionality

Description: Allows guests to perform a basic search by entering keywords (e.g., property name, location) in a search bar, retrieving relevant property listings based on keyword matches.

RequirementID: 5

Title: Advanced Search and Filtering

Description: Guests can filter search results based on various criteria, like price, pet-friendliness, and accessibility. This feature makes it easy for users to find properties that match their specific needs, saving time and improving their experience.

RequirementID: 6

Title: View Detailed Property Information

Description: Guests can view a property's full details, including high-quality images, rent, neighborhood characteristics. This reduces the need for initial inquiries and provides guests with a complete picture of what each property offers.

All Registered Users (Students,Landlords,Moderator,Admin)

All Users

RequirementID: 25

Title: User Account Creation

Description: Users can create accounts with specific roles and permissions, ensuring secure access to features.

RequirementID: 26

Title: User Login and Logout

Description: Users can securely log in to access the platform's features based on their role. Logging out protects account information, especially for students and landlords who may access sensitive messaging and profile data.

Students :

RequirementID: 1

Title: User Profile Management

Description: Students can complete or update their profiles with contact information, profile photos, and short bios. This helps personalize the user experience and builds credibility.

RequirementID: 4

Title: Basic Search Functionality

Description: Allows students to perform a basic search by entering keywords (e.g., property name, location) in a search bar, retrieving relevant property listings based on keyword matches.

RequirementID: 29

Title: Save Search Functionality

Description: This feature enables students to save their last 5 keyword searches as a "Search History" for convenience when they return to the platform. The saved searches will help users quickly access previous queries without the need to re-enter the keywords manually.

RequirementID: 5

Title: Advanced Search and Filtering

Description: Students can filter search results based on various criteria, like price, pet-friendliness, and accessibility. This feature makes it easy for users to find properties that match their specific needs, saving time and improving their experience.

RequirementID: 6

Title: View Detailed Property Information

Description: Students can view a property's full details, including high-quality images, rent, neighborhood characteristics, and landlord information. This reduces the need for initial inquiries and provides students with a complete picture of what each property offers.

RequirementID: 7

Title: In-Site Messaging System

Description: A secure, internal messaging system allows students and landlords to communicate directly on the platform. They can discuss property details, rental terms, and arrange viewings without relying on external communication, enhancing privacy and convenience.

RequirementID: 11

Title: Favorite Listings

Description: Students can save (favorite) listings of interest to revisit later, making it easy to keep track of potential rentals without performing multiple searches.This feature enhances convenience and helps students organize their property options.

RequirementID: 22

Title: Respond to Tenant Questionnaire

Description: Students can complete a landlord-provided questionnaire to showcase their suitability for a property. Their responses contribute to a calculated "Tenant Score" visible to landlords, enhancing their chances of securing a rental.

Landlords :

RequirementID: 1

Title: User Profile Management

Description: Landlords can complete or update their profiles with contact information, profile photos, and short bios. This helps personalize the user experience and builds credibility.

RequirementID: 2

Title: Creation of Property Listing

Description: Landlords can create detailed property listings by adding information such as rent, location, photos, property type, and amenities. This

ensures that students have the relevant details needed to evaluate a property without additional inquiries.

RequirementID: 3

Title: Editing of Property Listing

Description: Allows landlords to update existing listings to correct information or reflect changes, such as rent adjustments or new photos. This helps keep listings accurate and current, reducing misunderstandings with prospective renters.

RequirementID: 7

Title: In-Site Messaging System

Description: A secure, internal messaging system allows students and landlords to communicate directly on the platform. They can discuss property details, rental terms, and arrange viewings without relying on external communication, enhancing privacy and convenience.

RequirementID: 15

Title: Media Upload for Listings

Description: Landlords can upload media, allowing them to showcase different aspects of the property quickly and effectively. This feature ensures that listings are visually rich, providing students with a clearer view of the property.

RequirementID: 19

Title: Support for Multiple Property Listings per Landlord

Description: Landlords with more than one property can manage multiple listings under a single account, making it easy to track inquiries, update information, and maintain consistency across listings.

RequirementID: 22

Title: Tenant Score Assessment

Description: Landlords can fill a pre-defined questionnaire to evaluate prospective tenants. The system calculates a "Tenant Score" for each student based on their responses (checking how many answers match with the landlord's selections), providing an overall assessment to identify suitable renters.

Moderators :

RequirementID: 9

Title: Reviewing and Approving Property Listing

Description: The system shall enable moderators to review and approve property listings and user-generated content, ensuring compliance with platform guidelines before publication.

Priority 2 Requirements

Registered Users

Students :

RequirementID: 10

Title: Report Inappropriate Listings

Description: If students encounter inappropriate or misleading listings, they can report them to the moderator. This feedback is essential for maintaining a safe and accurate platform, as moderators can investigate reports and take corrective actions when needed.

RequirementID: 13

Title: Post-Rental Feedback and Ratings

Description: After renting a property, students can leave feedback and rate their experience with the landlord. This information is visible to other students, promoting transparency and helping future renters make informed choices based on past tenant experiences.

RequirementID: 17

Title: Privacy Settings for User Profiles

Description: Users can control which profile details are publicly visible, such as hiding contact information from unverified users. This feature gives users control over their privacy, improving security and ensuring a safe environment for both students and landlords.

RequirementID: 21

Title: Share Listings Externally

Description: Students can share listings via external channels, such as social media, email, or messaging apps, to increase visibility.

Landlords :

RequirementID: 16

Title: Automatic Listing Expiration and Renewal Notifications

Description: Listings have an automatic expiration period to ensure that only current rentals are displayed. Landlords receive notifications before expiry and can renew listings if the property is still available. This keeps the listing pool fresh and reduces outdated or inactive properties.

RequirementID: 17

Title: Privacy Settings for User Profiles

Description: Users can control which profile details are publicly visible, such as hiding contact information from unverified users. This feature gives users control over their privacy, improving security and ensuring a safe environment for both students and landlords.

RequirementID: 21

Title: Share Listings Externally

Description: Landlords can share listings via external channels, such as social media, email, or messaging apps, to increase visibility.

RequirementID: 30

Title: Tenant Score Assessment - Advanced Implementation

Description: Landlords can create a custom questionnaire with weighted questions to evaluate prospective tenants. The system calculates a "Tenant Score" for each student based on their responses, providing an overall assessment to identify suitable renters.

Moderators :

RequirementID: 31

Title: Moderator Dashboard for Analytics

Description: Moderators can access analytics, including most favorited properties , most applied properties , and most visited properties to monitor trends and also see the number of registered users based on their role.

Admin :

RequirementID: 8

Title: Admin Dashboard for Analytics and Moderation

Description: Admins can access site analytics, including user activity, login attempts, and page views, to monitor trends and detect unusual patterns. They can also review and address flagged content submitted by moderators.

Priority 3 Requirements

Registered Users

Students :

RequirementID: 14

Title: Saved Searches with Alerts

Description: Students can save their search preferences and get notified when a new property matches their criteria. This way, they don't have to keep searching manually, making the rental process easier and faster.

RequirementID: 18

Title: In-Platform Document Exchange

Description: Students can securely upload and exchange necessary documents (e.g., rental agreements) directly on the platform. This keeps document handling organized and confidential, enhancing the rental process's security and convenience.

RequirementID: 23

Title: Display Nearby Amenities

Description: The system shall display nearby amenities such as cafes, transport hubs, grocery stores, and other points of interest when a student views a property's "Nearby Amenities" feature.

RequirementID: 27

Title: Virtual Property Tours

Description: Students can take 360-degree virtual tours of properties, allowing them to view spaces remotely. This feature is especially helpful for international students or those unable to visit in person, providing a realistic view of the property.

RequirementID: 28

Title: Video Call Request Between Student and Landlord

Description: The platform enables students and landlords to request and schedule video calls within the messaging system, allowing them to discuss rental details directly. Both parties can suggest or confirm time slots

Landlords :

RequirementID: 12

Title: Creating a Draft Listing

Description: Landlords can create a property listing as a draft before publishing. This allows them to refine details over time without committing the listing publicly. They can review and finalize information, ensuring accuracy before making it visible to students.

RequirementID: 18

Title: In-Platform Document Exchange

Description: Landlords can securely upload and exchange necessary documents (e.g., rental agreements) directly on the platform. This keeps document handling organized and confidential, enhancing the rental process's security and convenience.

RequirementID: 20

Title: Badges for Verified Landlords

Description: Verified landlords receive a badge on their profile, indicating they have passed a verification process. This badge helps students identify trusted landlords, promoting a safer and more reliable rental marketplace.

RequirementID: 28

Title: Video Call Request Between Student and Landlord

Description: The platform enables students and landlords to request and schedule video calls within the messaging system, allowing them to discuss rental details directly. Both parties can suggest or confirm time slots

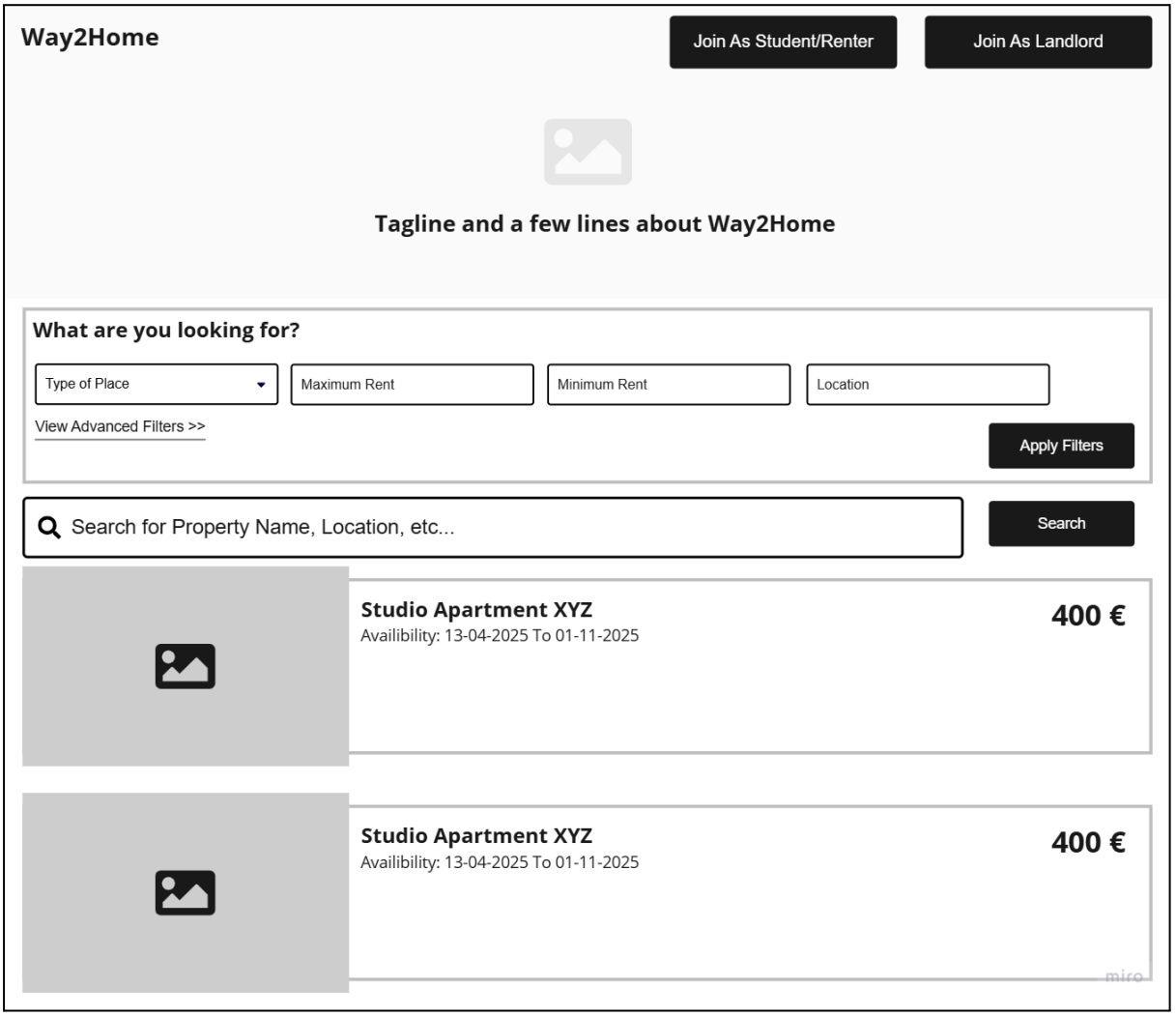
9. List of main data items and entities:

Data Item	Main Attributes	Description
User	● Name ● Role ● Avatar (profile photo) ● About ● Gender ● Birth date ● Contact Information	Represents the users of the platform, each with a role (Admin, Moderator, Landlord, or Student) and personal details.
Property	● Rent ● Deposit ● Location ● Property type ● Amenities ● Media files ● Description ● Availability ● Approval Status (Pending, Approved, Rejected) ● Size	Represents the properties listed by landlords, including key details like price, location, and features.
Amenity	● Name ● Description	Represents a feature of a property. E.g. WiFi, garage, etc.
Flagged Property	● Reporter ● Reason	Represents flagged properties, ensuring they are reviewed and moderated.
Saved Search	● Query ● Filters	Represents the criteria students use to search for properties, such as location, price, and amenities. This will help to

		personalize alerts to notify users of new listings matching their criteria.
Favorite Property	● Property	Represents properties that students save for future reference.
Questionnaire	● Questions ● Answers ● Tenant score	Represents a sheet of questions & answers on how a tenant has experienced the property. The score is calculated based on the answers.
Review	● Rating ● Review Text ● Creation date	Represents ratings and comments given by students about properties.
Media file	● Media type ● S3 id	Represents media files like images, videos and documents.
Message	● Content ● Creation date ● Media files	Represents messages exchanged between landlords and students, including the content and timestamp.

10. UI-Mockups and Storyboards

Storyboard 1:
Michael, a guest user, explores property listings to view basic information such as rent, location, and photos on the listings page without the need for an account, with limited platform interaction. He Doesn't have any option to Favorite the listing and cannot contact the landlord but is free to browse the listings


Way2Home
Join As Student/Renter
Join As Landlord
Tagline and a few lines about Way2Home
What are you looking for?
Type of Place
Maximum Rent
Minimum Rent
Location
View Advanced Filters >>
Apply Filters
Search for Property Name, Location, etc...
Search
Studio Apartment XYZ
Availibility: 13-04-2025 To 01-11-2025
400 €
Studio Apartment XYZ
Availibility: 13-04-2025 To 01-11-2025
400 €
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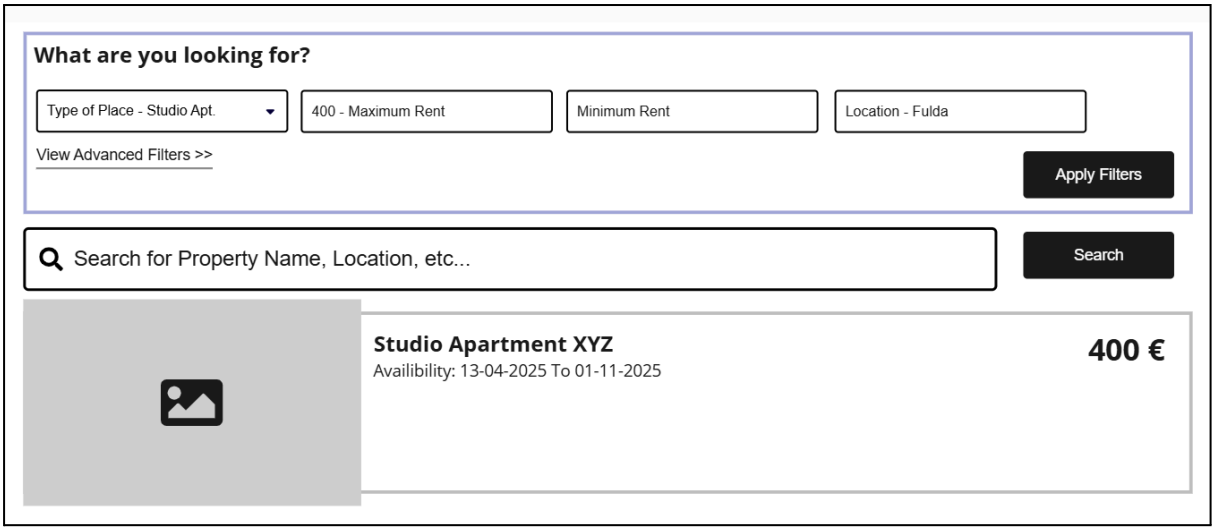
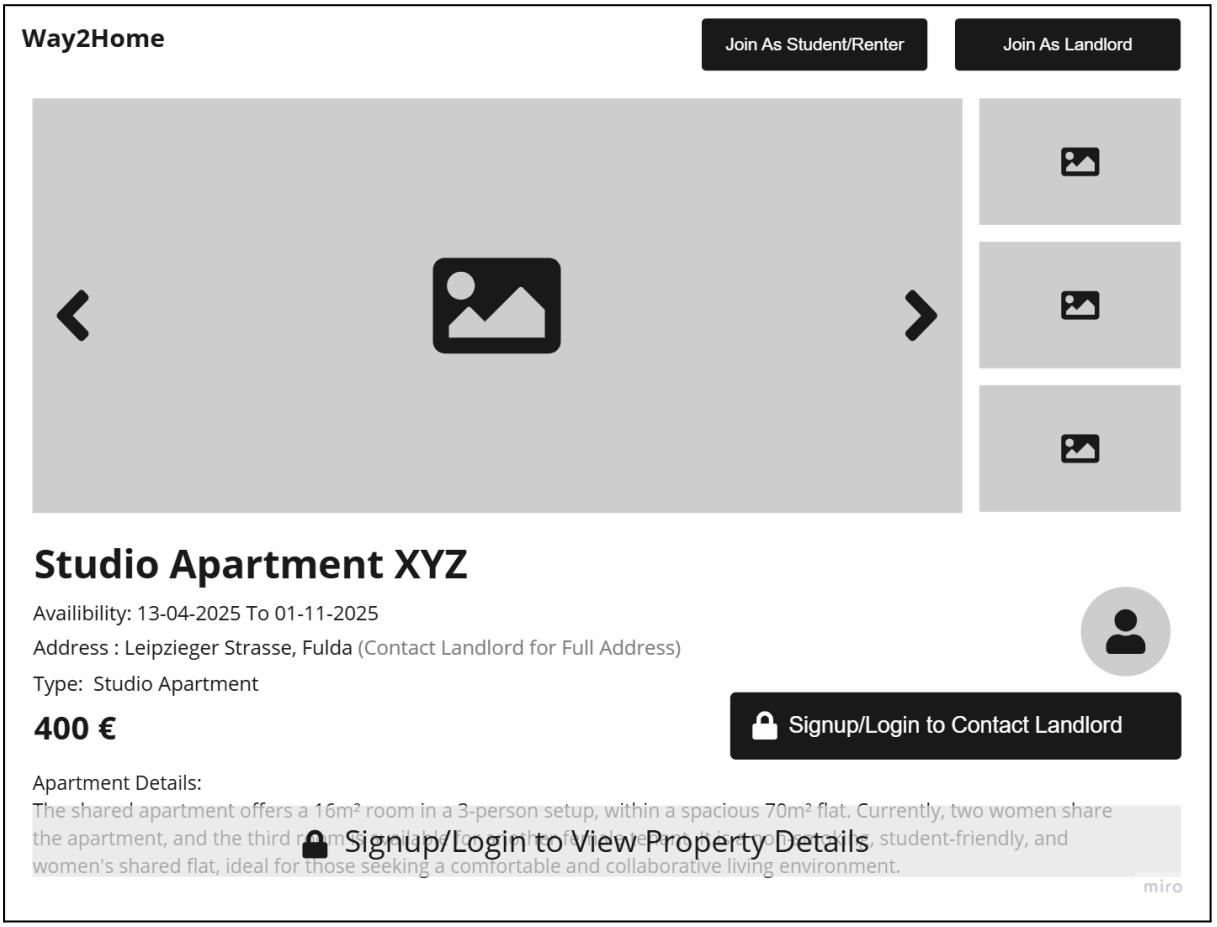
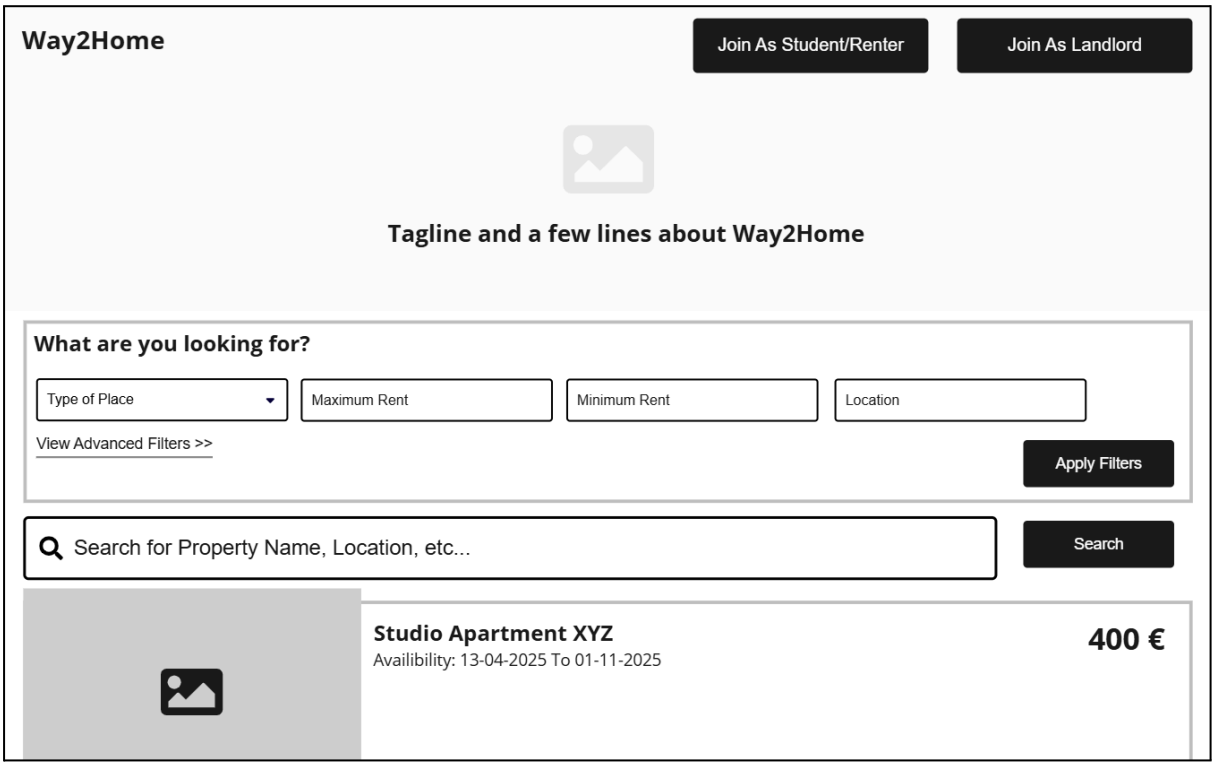
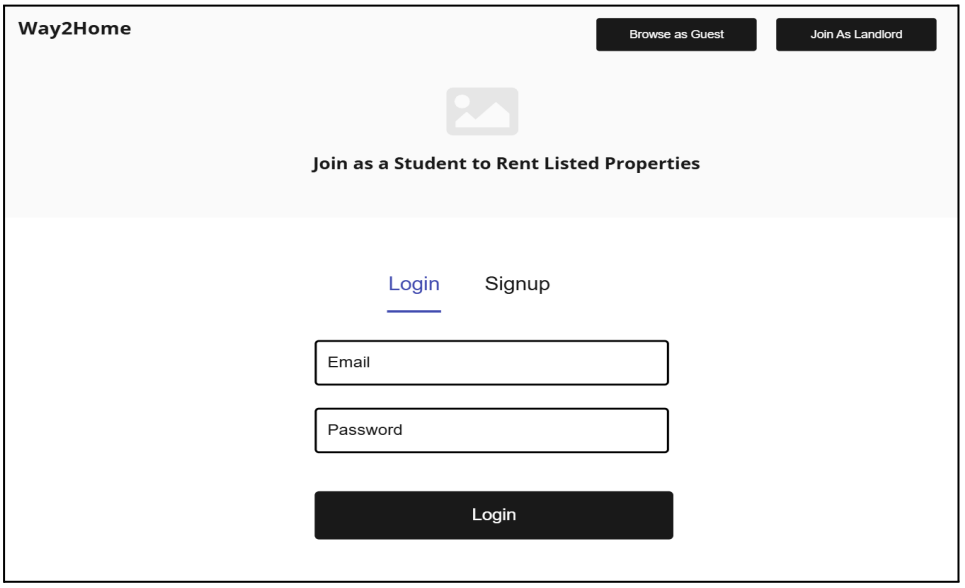
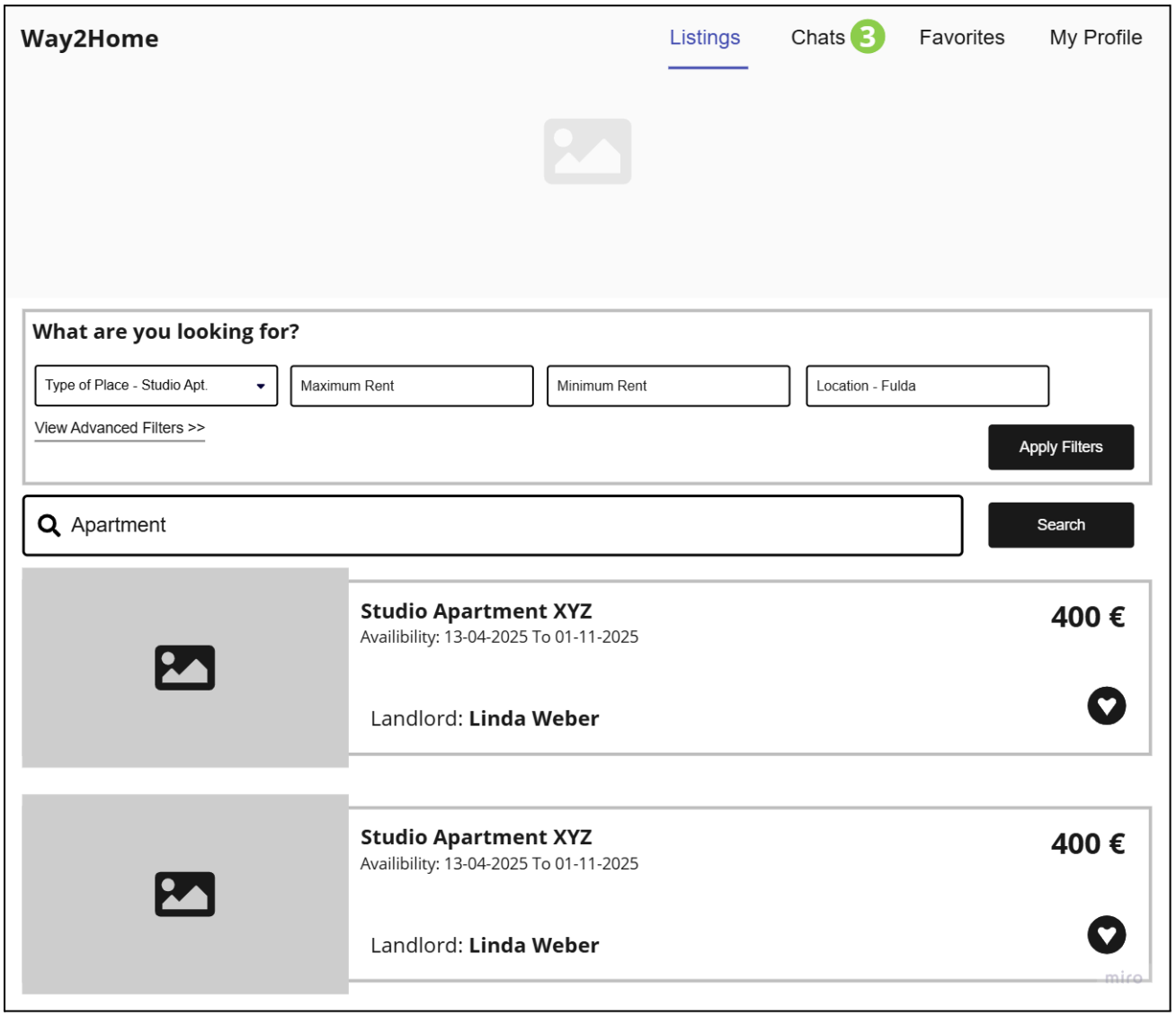

What are you looking for?
Type of Place - Studio Apt.
400 - Maximum Rent
Minimum Rent
Location - Fulda
View Advanced Filters >>
Apply Filters
Search for Property Name, Location, etc...
Search
Studio Apartment XYZ
Availibility: 13-04-2025 To 01-11-2025
400 €

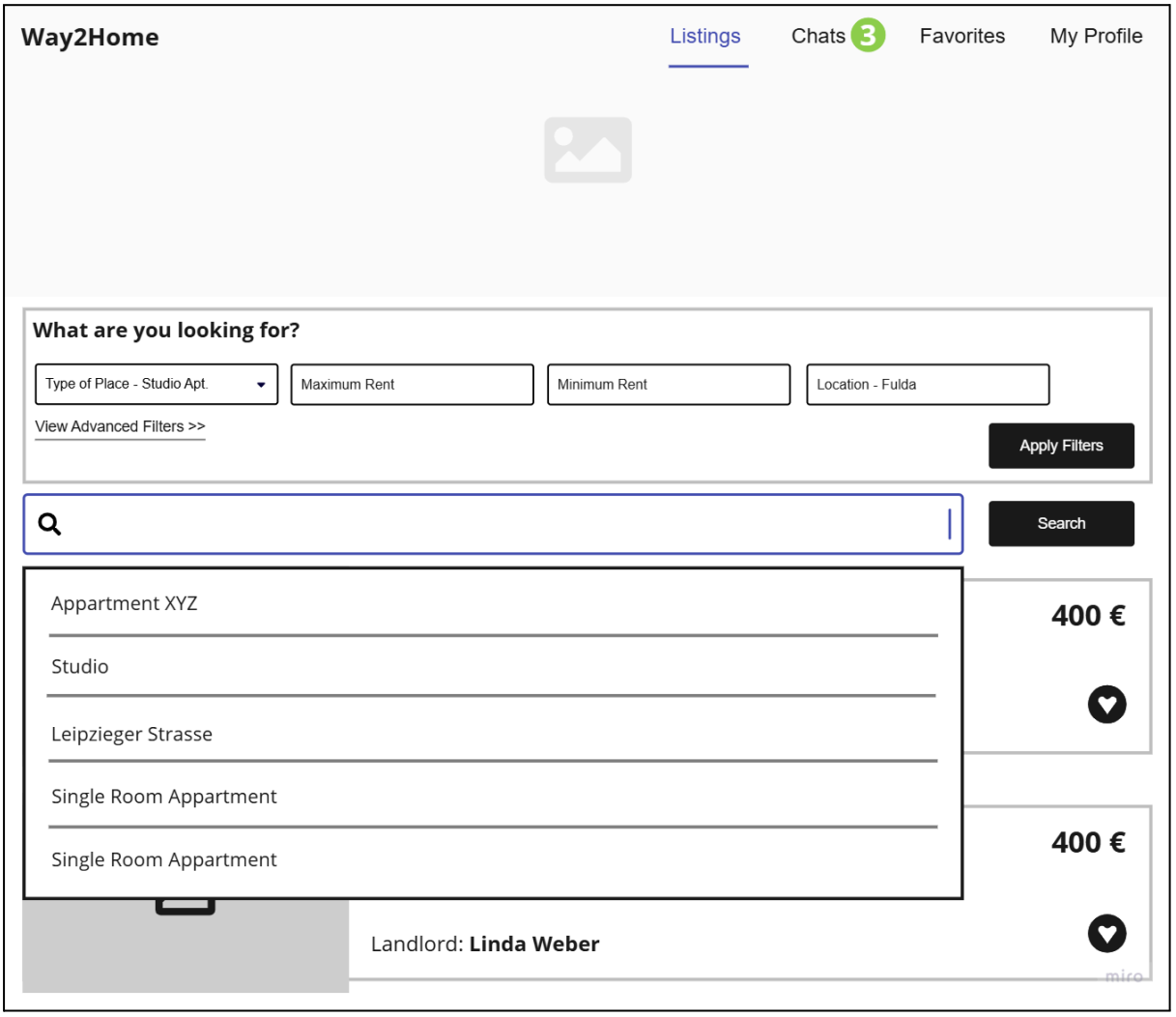


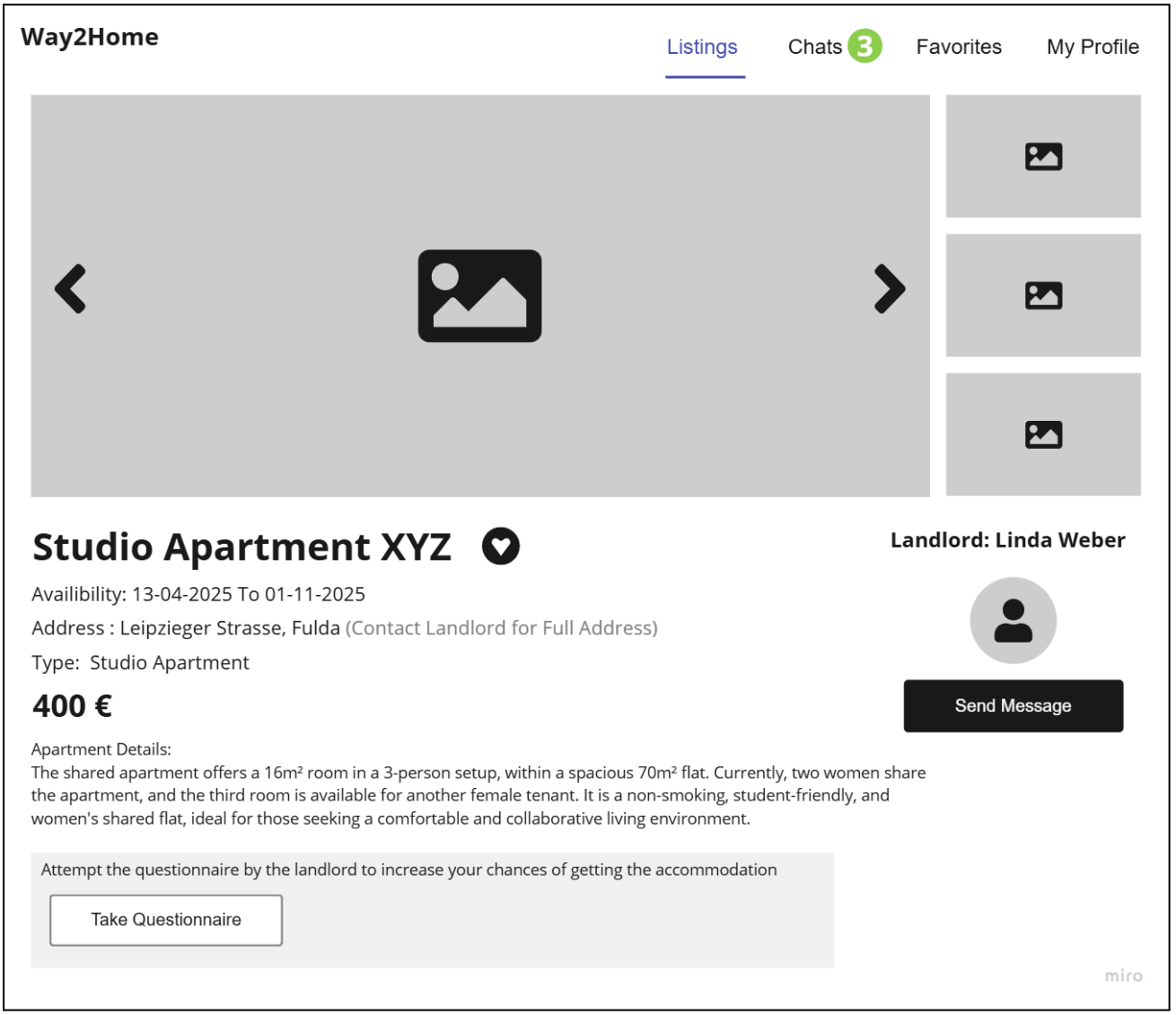
Storyboard 2:
Emily, a student, logs into her account by going to the Join as Student/Renter option to browse and search for housing options. She initially views unfiltered listings and then uses filters to narrow her search by selecting an apartment with a maximum rent of 400, refining her results to match her budget constraints. Then views the details of the properties displayed. She can add the listings as favorites, has saved searches and can also attempt questionnaires against the Property.


Way2Home
Join As Student/Renter
Join As Landlord
Tagline and a few lines about Way2Home
What are you looking for?
Type of Place
Maximum Rent
Minimum Rent
Location
View Advanced Filters >>
Apply Filters
Search for Property Name, Location, etc...
Search
Studio Apartment XYZ
Availibility: 13-04-2025 To 01-11-2025
400 €


Way2Home
Browse as Guest
Join As Landlord
Join as a Student to Rent Listed Properties
Login
Signup
Email
Password
Login


Way2Home
Listings
Chats 3
Favorites
My Profile
What are you looking for?
Type of Place - Studio Apt.
Maximum Rent
Minimum Rent
Location - Fulda
View Advanced Filters >>
Apply Filters
Apartment
Search
Studio Apartment XYZ
Availibility: 13-04-2025 To 01-11-2025
400 €
Landlord: Linda Weber
Studio Apartment XYZ
Availibility: 13-04-2025 To 01-11-2025
400 €
Landlord: Linda Weber
miro


Way2Home
Listings
Chats 3
Favorites
My Profile
What are you looking for?
Type of Place - Studio Apt.
Maximum Rent
Minimum Rent
Location - Fulda
View Advanced Filters >>
Apply Filters
Search
Appartment XYZ
Studio
Leipzieger Strasse
Single Room Appartment
Single Room Appartment
400 €
400 €
Landlord: Linda Weber
miro


Way2Home
Listings
Chats 3
Favorites
My Profile
Studio Apartment XYZ
Landlord: Linda Weber
Availibility: 13-04-2025 To 01-11-2025
Address : Leipzieger Strasse, Fulda (Contact Landlord for Full Address)
Type: Studio Apartment
400 €
Send Message
Apartment Details:
The shared apartment offers a 16m² room in a 3-person setup, within a spacious 70m² flat. Currently, two women share the apartment, and the third room is available for another female tenant. It is a non-smoking, student-friendly, and women's shared flat, ideal for those seeking a comfortable and collaborative living environment.
Attempt the questionnaire by the landlord to increase your chances of getting the accommodation
Take Questionnaire
miro

Way2Home

Listings | Chats 3 | Favorites | My Profile

Studio Apartment XYZ - Questionnaire

What is your preferred lease duration?

- [ ] 6 months
- [x] 12 months
- [ ] 24 months or more
- [ ] Month-to-month

How often do you host guests?

- [ ] Rarely or never
- [ ] Occasionally (1 - 2
- [x] Frequently (every weekend)
- [ ] Long Term Visitors

What is your current employment status?

- [ ] Part-time employed
- [x] Self-employed
- [ ] Unemployed
- [ ] Blocked amount

Do you have any pets?

- [x] No pets
- [ ] Yes, (e.g., cat, small
- [ ] Yes, multiple pets
- [ ] Yes, (exotic animals)

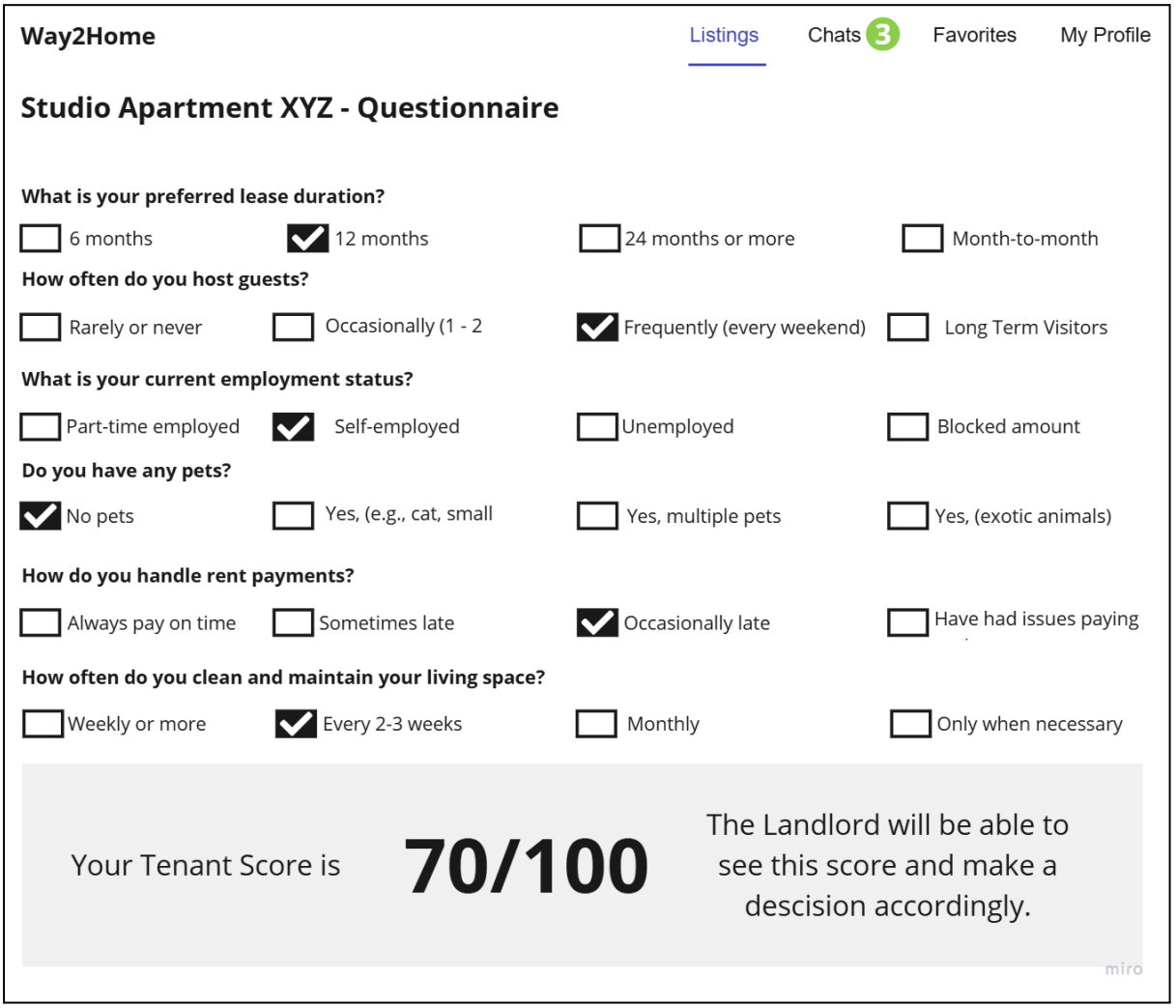
How do you handle rent payments?

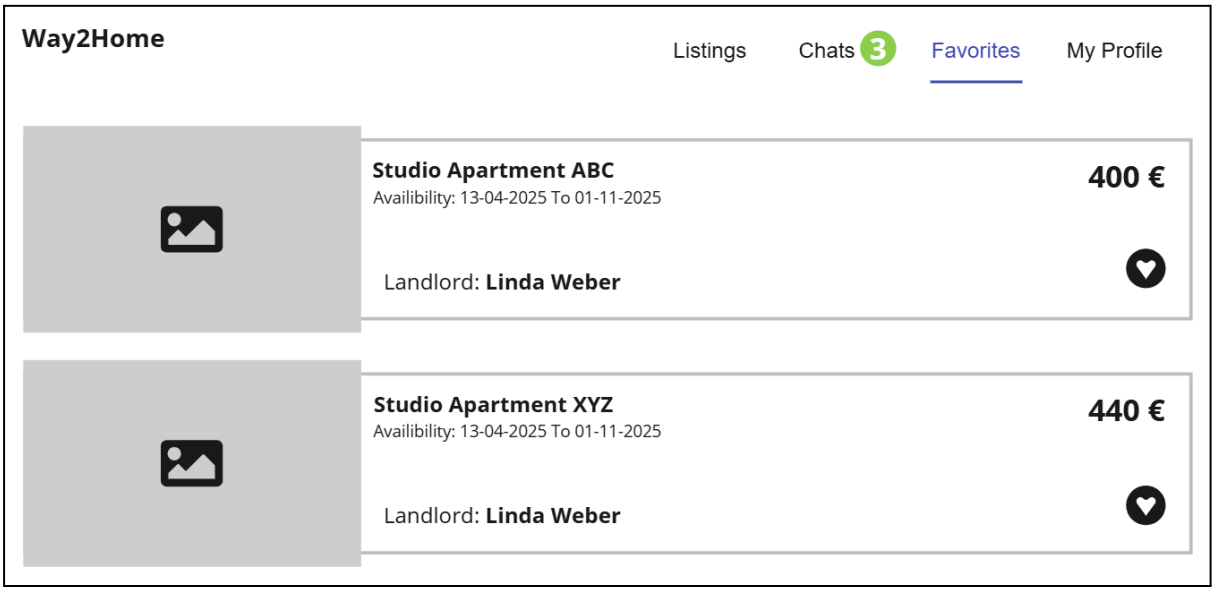
- [ ] Always pay on time
- [ ] Sometimes late
- [x] Occasionally late
- [ ] Have had issues paying

How often do you clean and maintain your living space?

- [ ] Weekly or more
- [x] Every 2-3 weeks
- [ ] Monthly
- [ ] Only when necessary

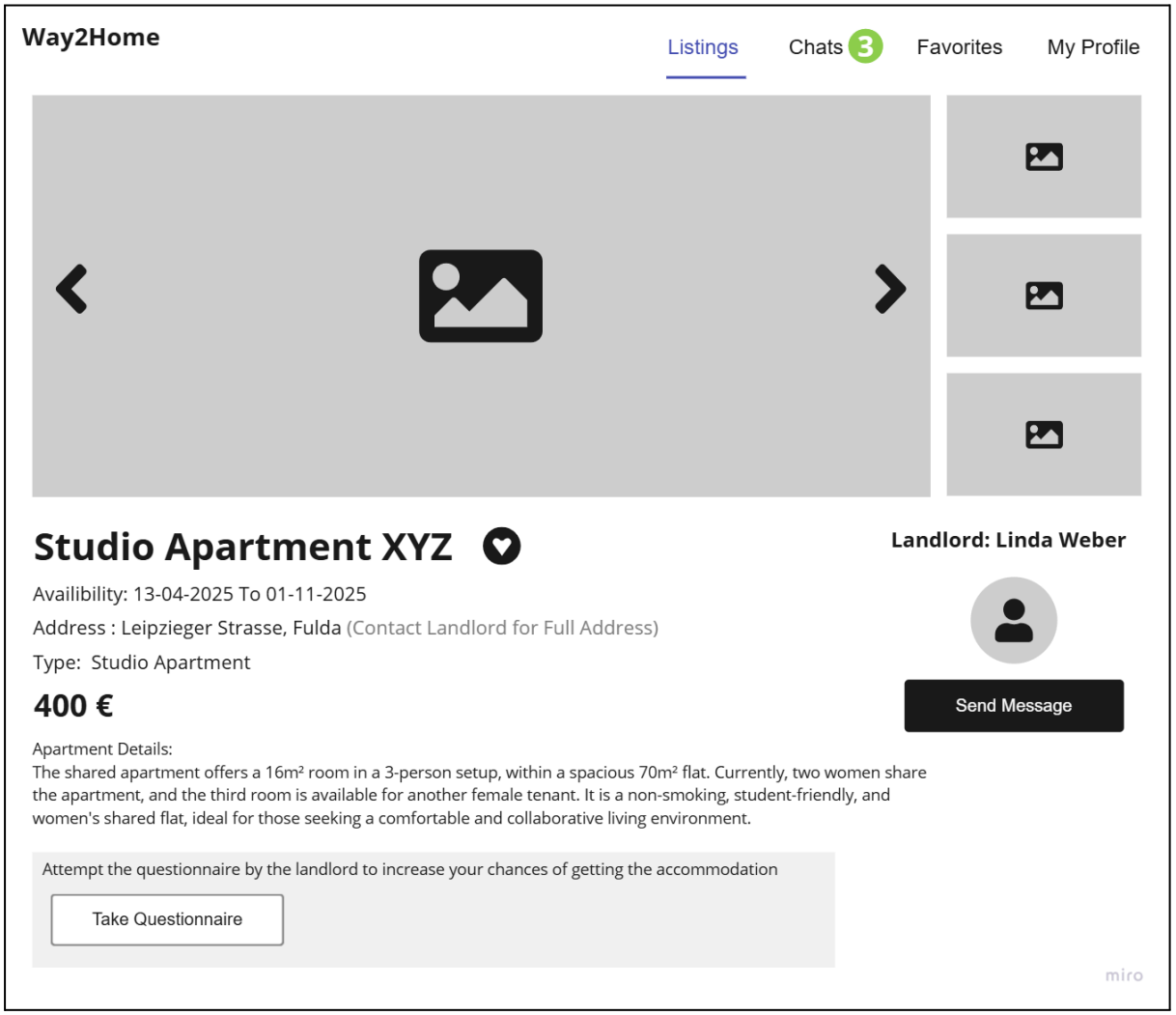
Submit | Back

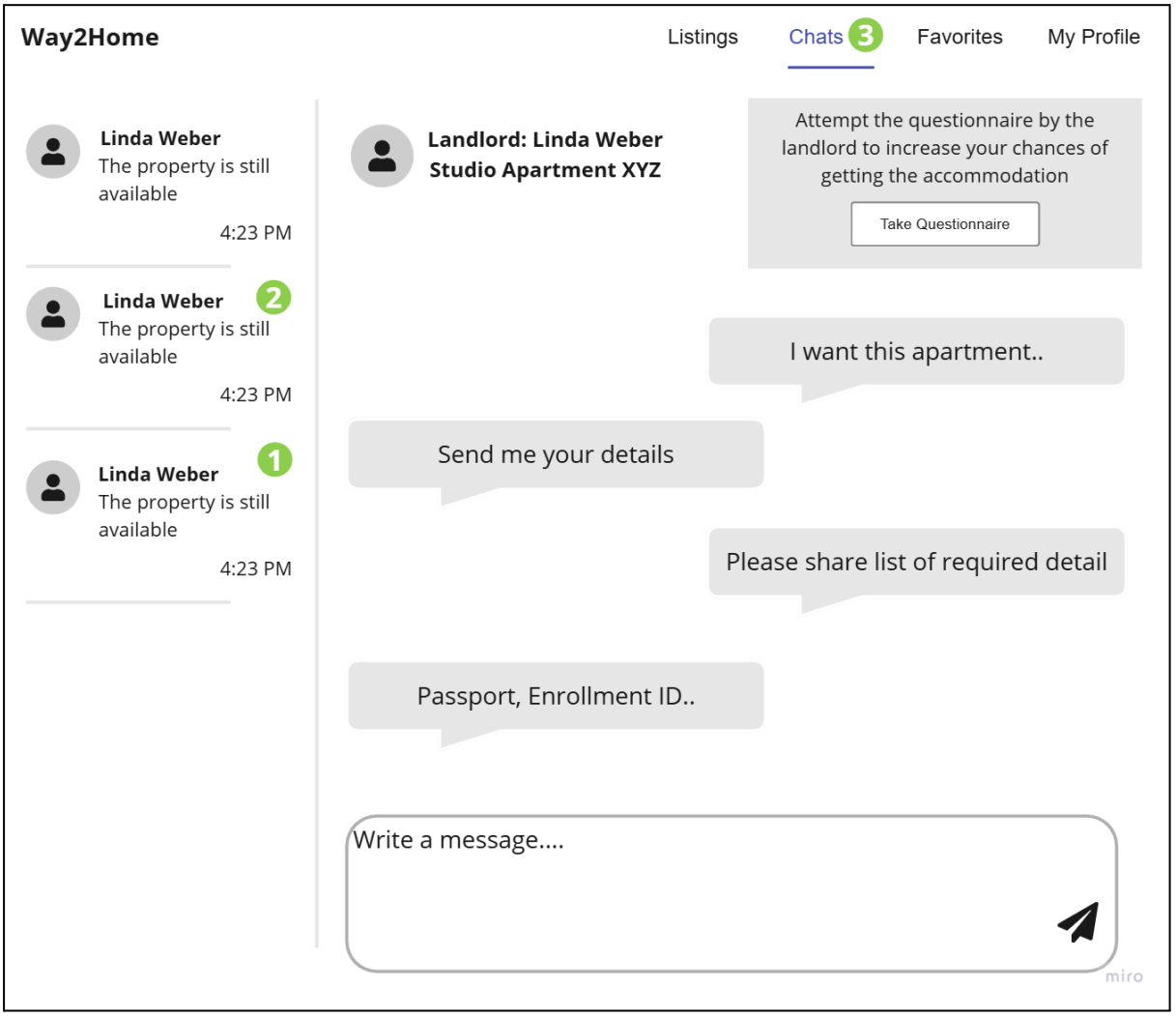

Way2Home
Listings
Chats 3
Favorites
My Profile
Studio Apartment XYZ - Questionnaire
What is your preferred lease duration?
6 months
12 months
24 months or more
Month-to-month
How often do you host guests?
Rarely or never
Occasionally (1 - 2
Frequently (every weekend)
Long Term Visitors
What is your current employment status?
Part-time employed
Self-employed
Unemployed
Blocked amount
Do you have any pets?
No pets
Yes, (e.g., cat, small
Yes, multiple pets
Yes, (exotic animals)
How do you handle rent payments?
Always pay on time
Sometimes late
Occasionally late
Have had issues paying
How often do you clean and maintain your living space?
Weekly or more
Every 2-3 weeks
Monthly
Only when necessary
Your Tenant Score is
70/100
The Landlord will be able to see this score and make a descision accordingly.
miro

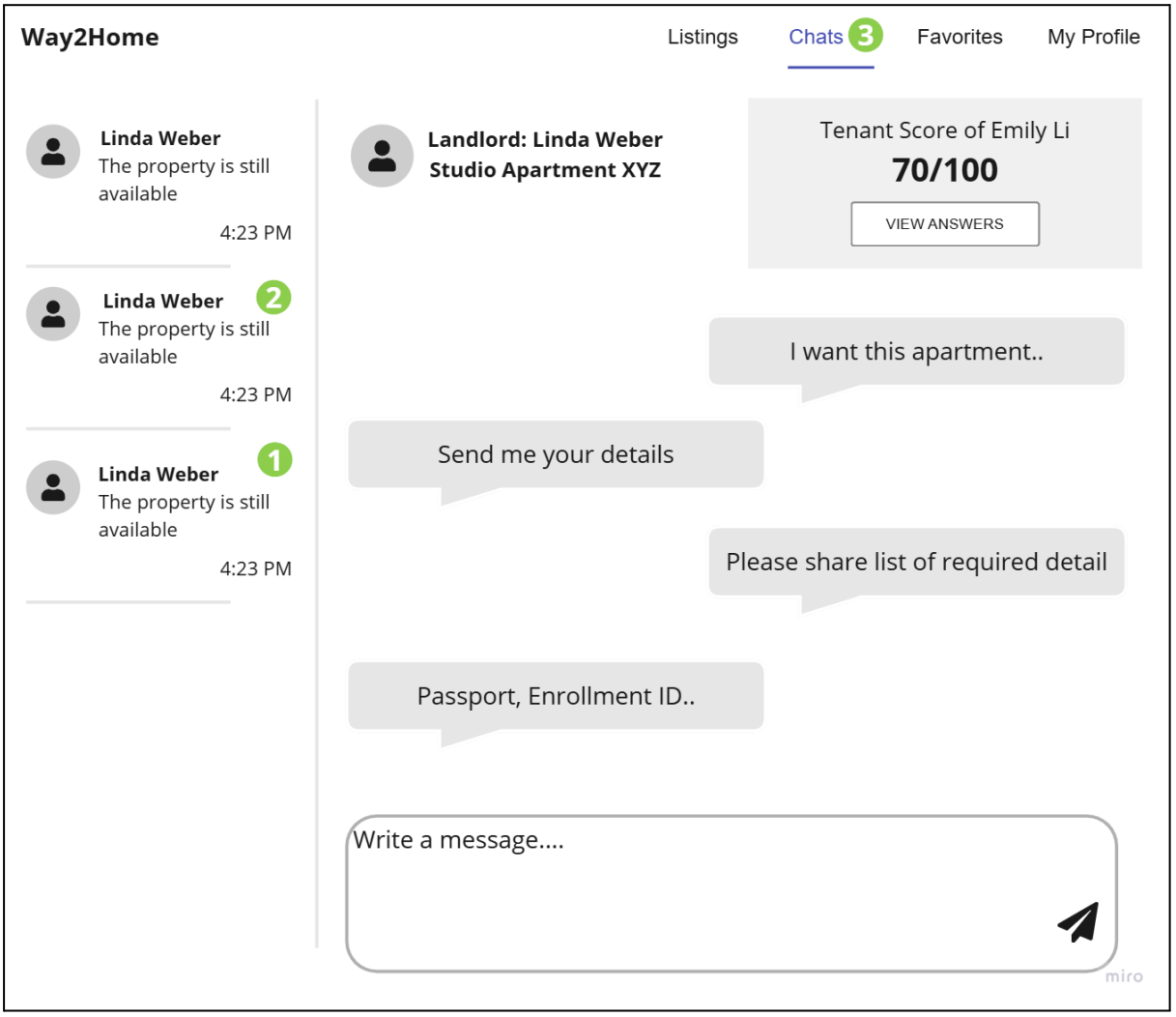

Way2Home
Listings
Chats 3
Favorites
My Profile
Studio Apartment ABC
Availibility: 13-04-2025 To 01-11-2025
400 €
Landlord: Linda Weber
Studio Apartment XYZ
Availibility: 13-04-2025 To 01-11-2025
440 €
Landlord: Linda Weber

Storyboard 3:

Emily, as a logged in student, utilizes the in-app messaging feature to directly contact Linda, the landlord, with questions about a property's lease terms and pet policy, facilitating secure and convenient communication within the platform.

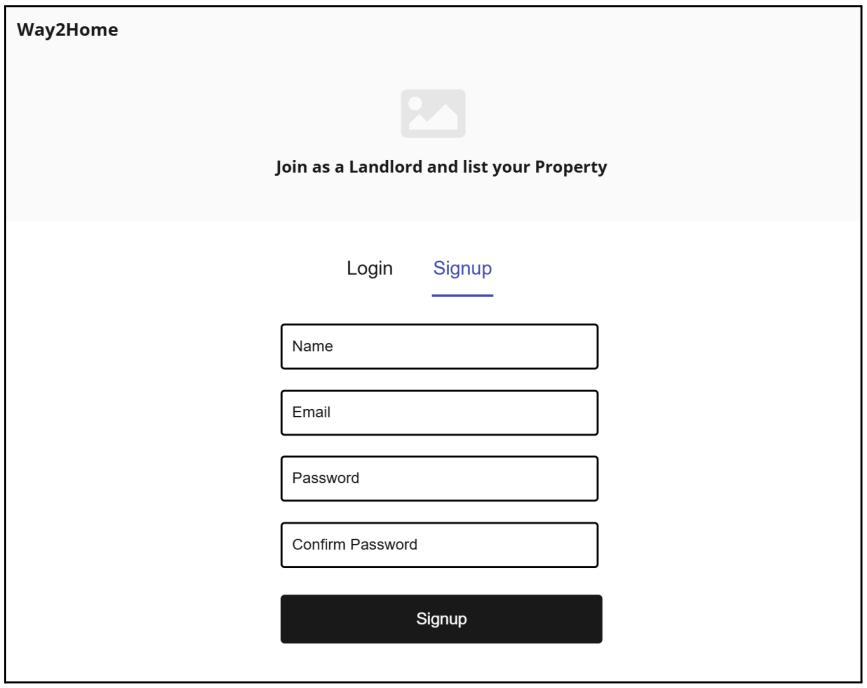


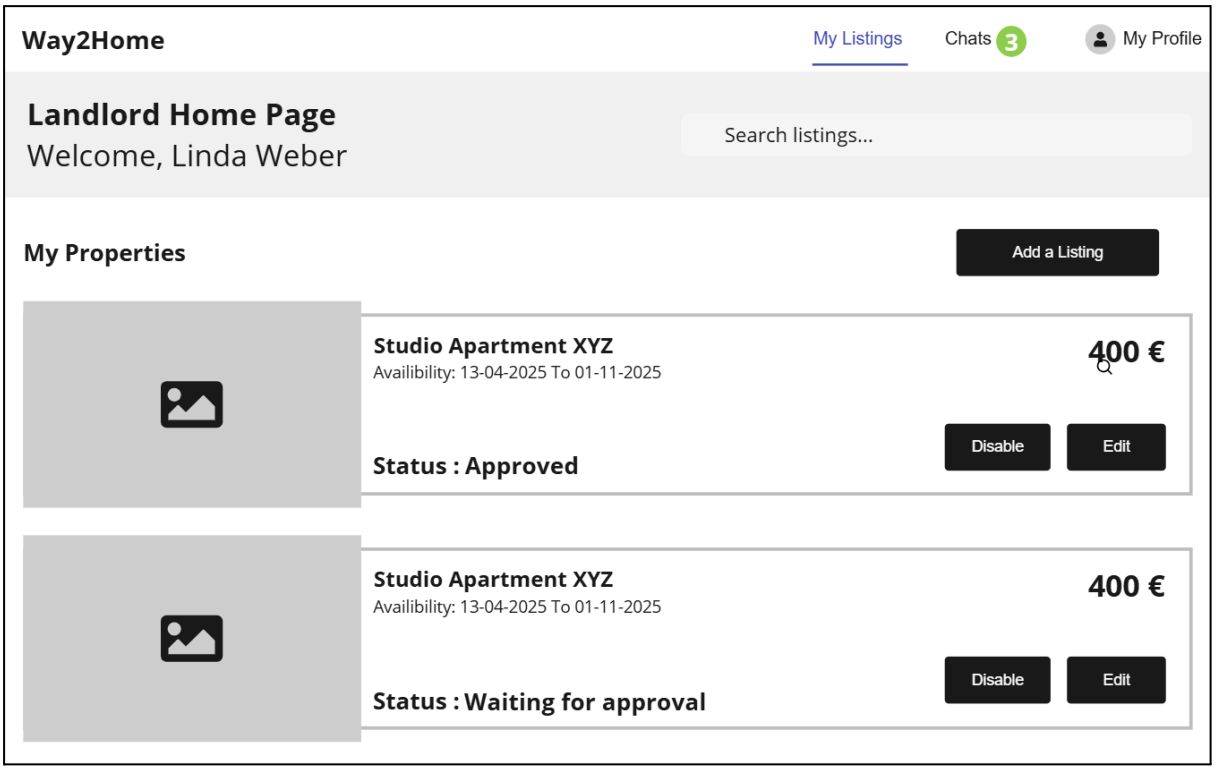

Way2Home
Listings
Chats 3
Favorites
My Profile
Linda Weber
The property is still available
4:23 PM
Linda Weber 2
The property is still available
4:23 PM
Linda Weber 1
The property is still available
4:23 PM
Landlord: Linda Weber
Studio Apartment XYZ
Attempt the questionnaire by the landlord to increase your chances of getting the accommodation
Take Questionnaire
I want this apartment..
Send me your details
Please share list of required detail
Passport, Enrollment ID..
Write a message....
miro

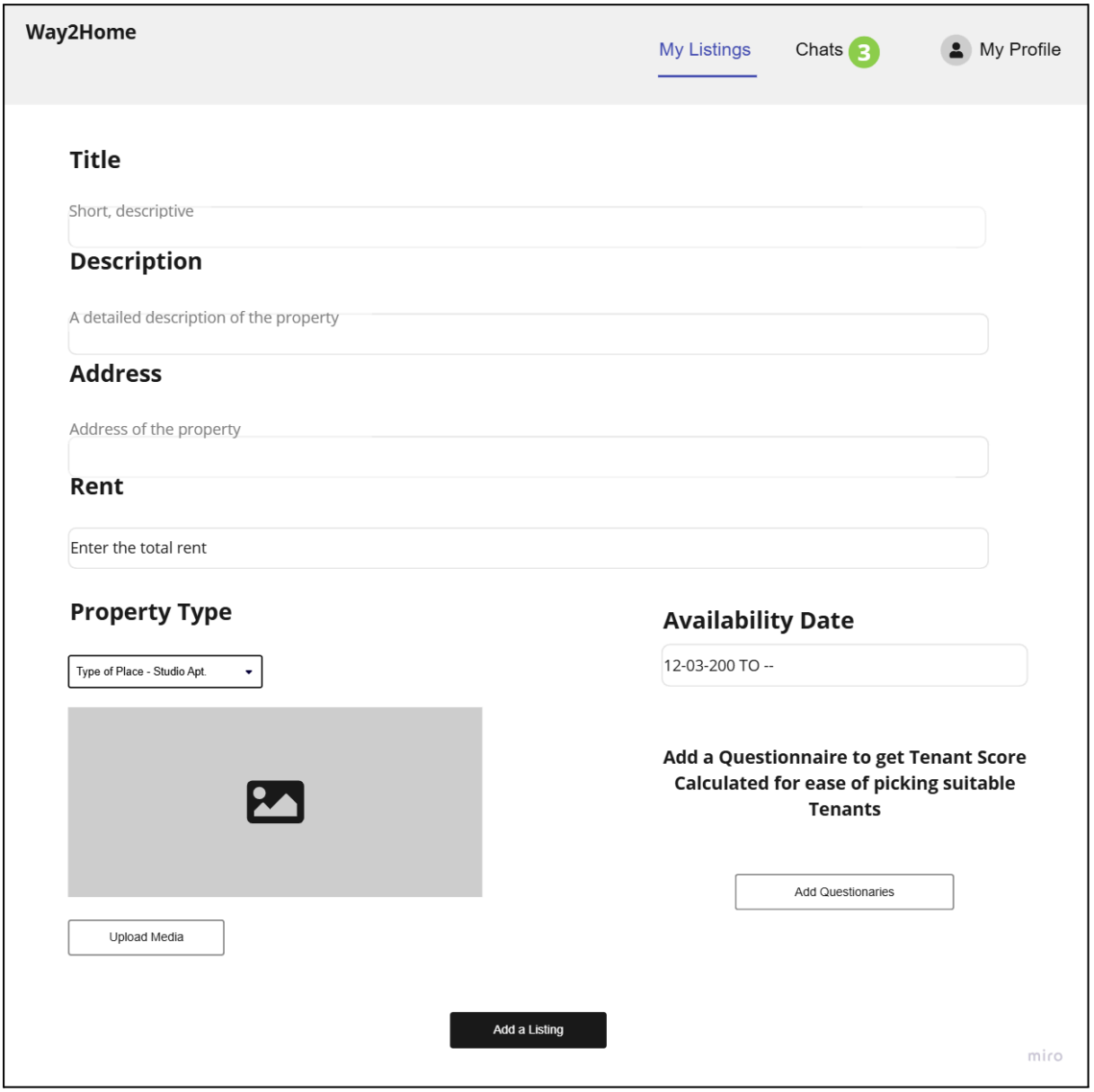

Way2Home
Listings
Chats 3
Favorites
My Profile
Linda Weber
The property is still available
4:23 PM
Linda Weber 2
The property is still available
4:23 PM
Linda Weber 1
The property is still available
4:23 PM
Landlord: Linda Weber
Studio Apartment XYZ
Tenant Score of Emily Li
70/100
VIEW ANSWERS
I want this apartment..
Send me your details
Please share list of required detail
Passport, Enrollment ID..
Write a message....
miro

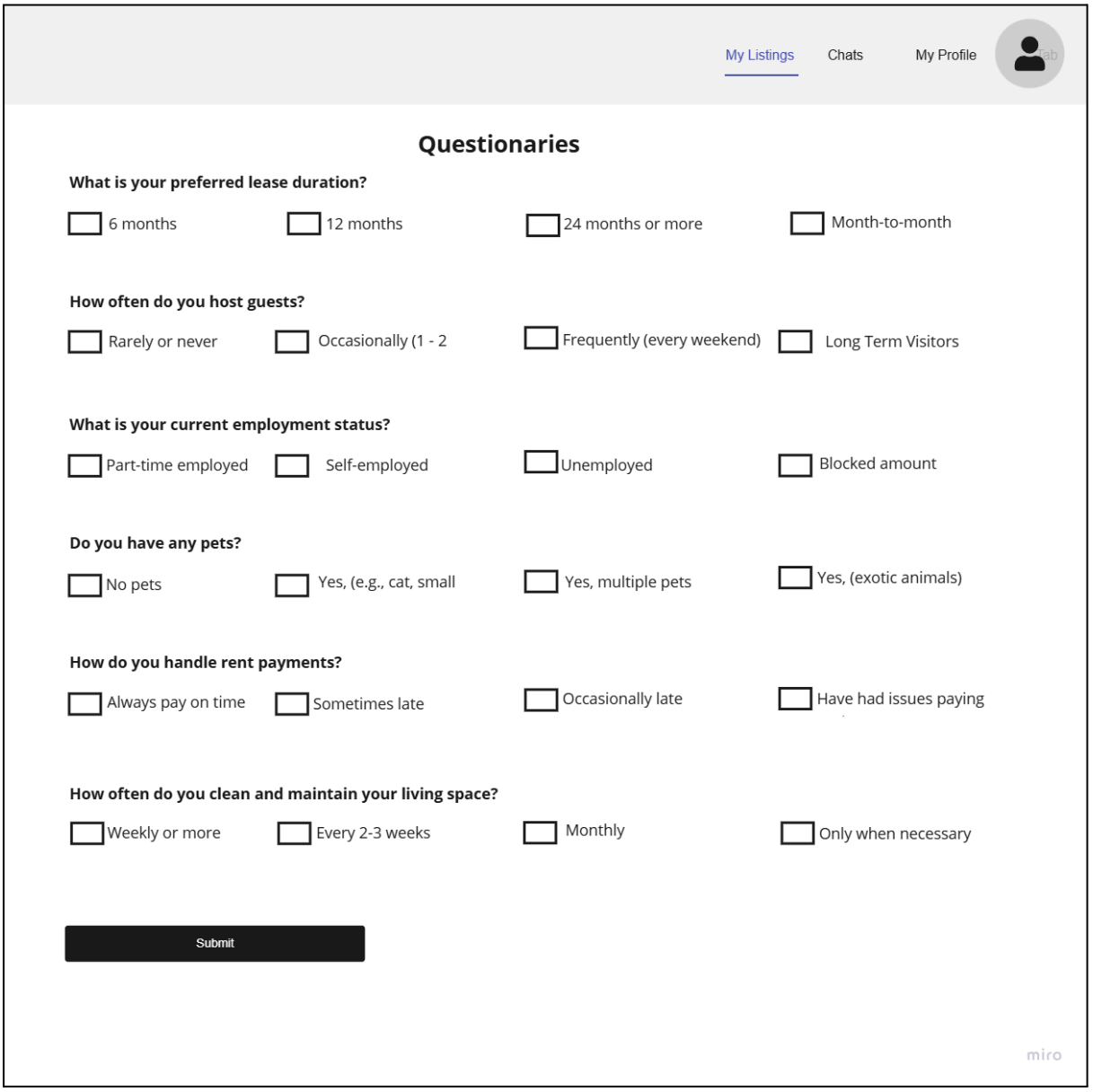
Storyboard 4:

Linda selects the "Join As Landlord" option on the landing page, proceeds to create an account, and prepares to post a new property listing. She provides detailed information, uploads photos, and includes additional features to appeal to potential tenants. Her submission undergoes a review process before being published on the platform.



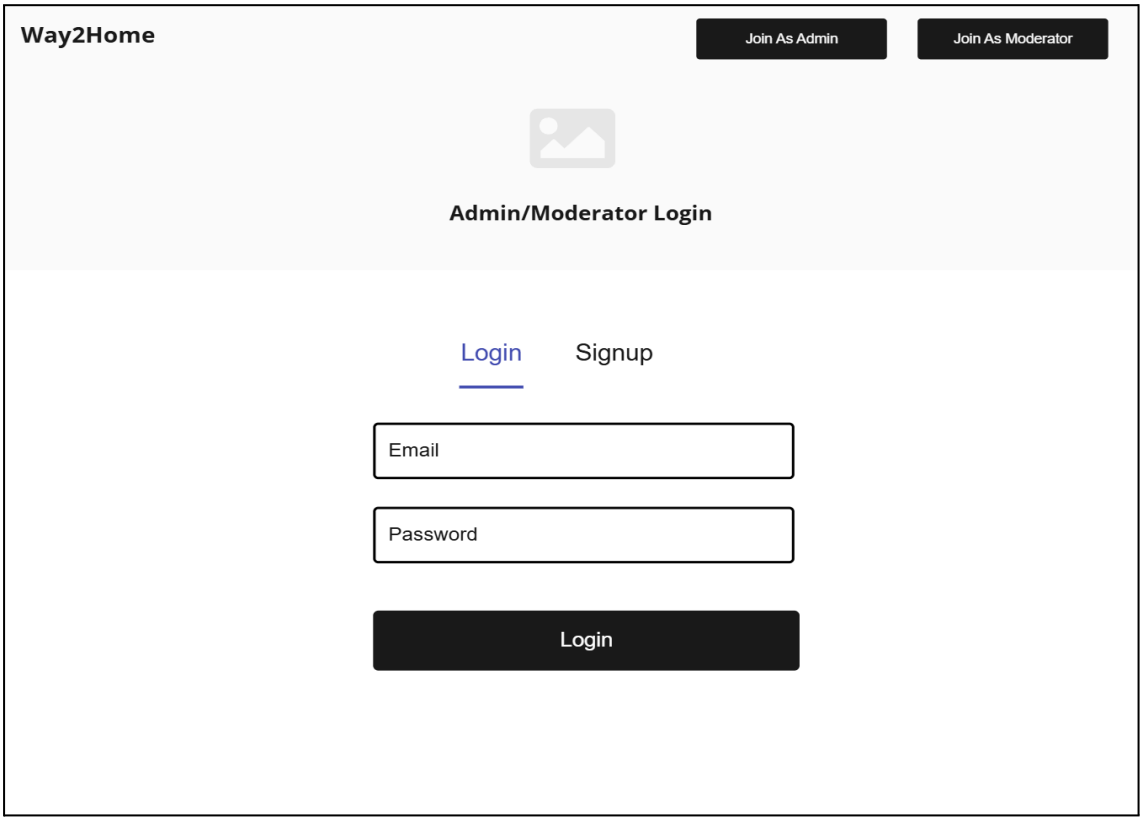


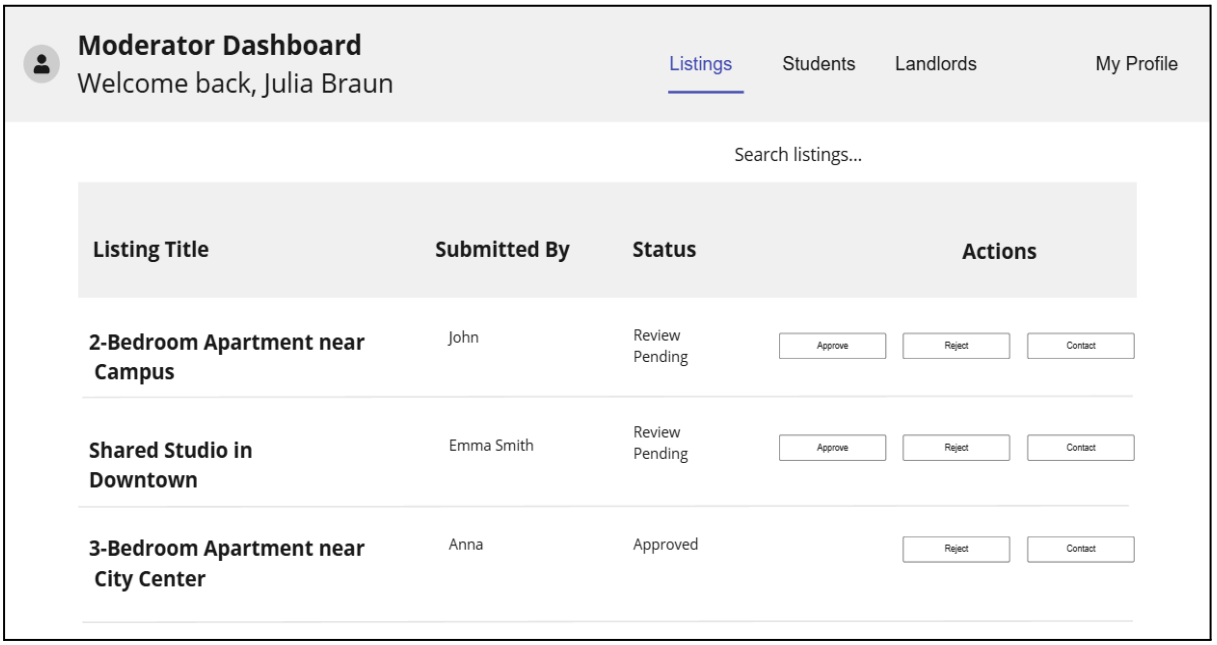

Way2Home
My Listings
Chats 3
My Profile
Title
Short, descriptive
Description
A detailed description of the property
Address
Address of the property
Rent
Enter the total rent
Property Type
Type of Place - Studio Apt.
Availability Date
12-03-200 TO --
Add a Questionnaire to get Tenant Score Calculated for ease of picking suitable Tenants
Add Questionaries
Upload Media
Add a Listing
miro

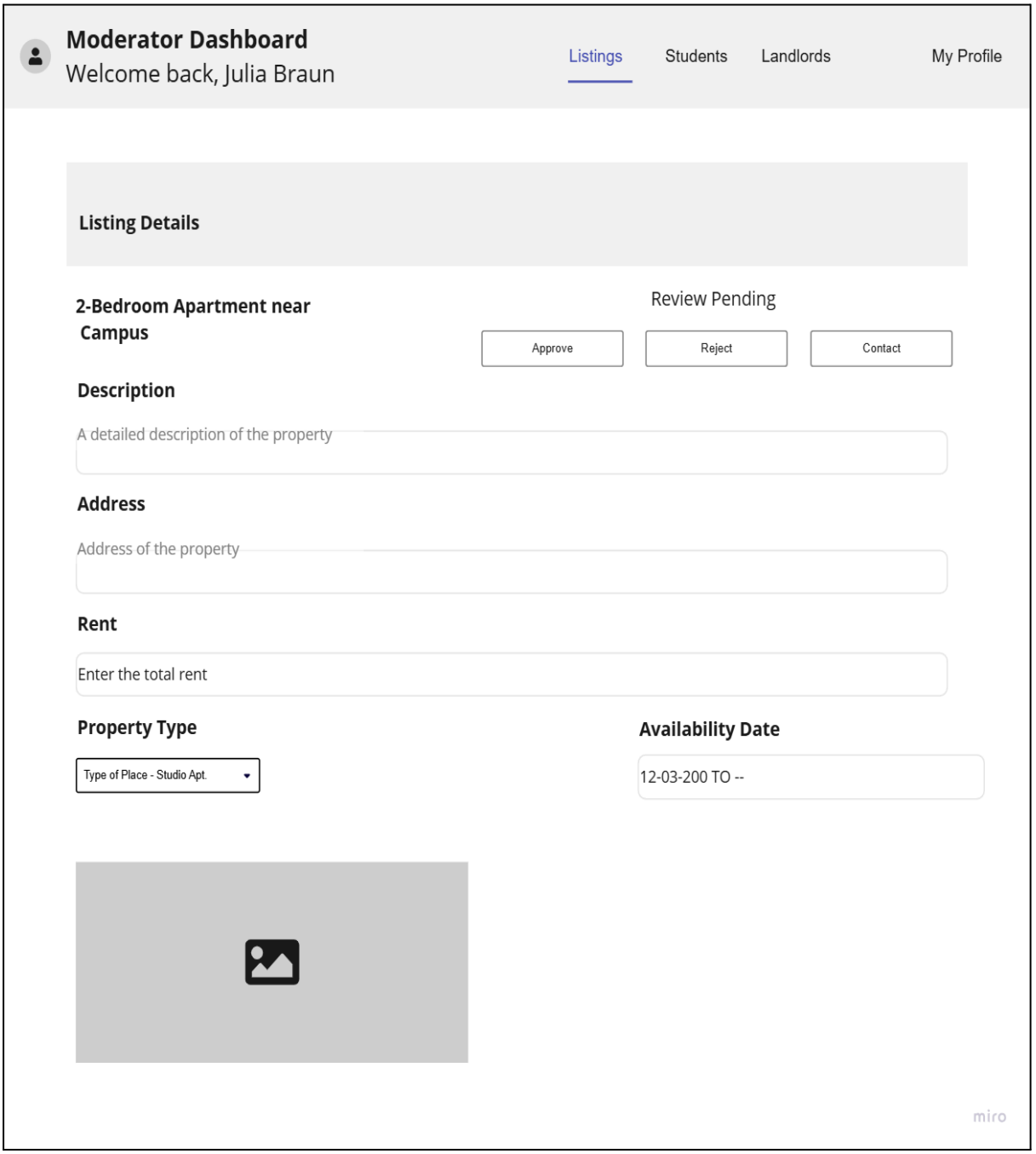

My Listings
Chats
My Profile
Questionaries
What is your preferred lease duration?
6 months
12 months
24 months or more
Month-to-month
How often do you host guests?
Rarely or never
Occasionally (1 - 2
Frequently (every weekend)
Long Term Visitors
What is your current employment status?
Part-time employed
Self-employed
Unemployed
Blocked amount
Do you have any pets?
No pets
Yes, (e.g., cat, small
Yes, multiple pets
Yes, (exotic animals)
How do you handle rent payments?
Always pay on time
Sometimes late
Occasionally late
Have had issues paying
How often do you clean and maintain your living space?
Weekly or more
Every 2-3 weeks
Monthly
Only when necessary
Submit
miro

Storyboard 5:

Julia logs into the admin panel with her moderator role, which provides her with limited access tailored to her responsibilities for reviewing and approving property listings. Julia evaluates Linda's property listing for quality, clarity, and adherence to platform policies. She provides feedback for necessary changes and, once satisfied, approves the listing for publication on the website.

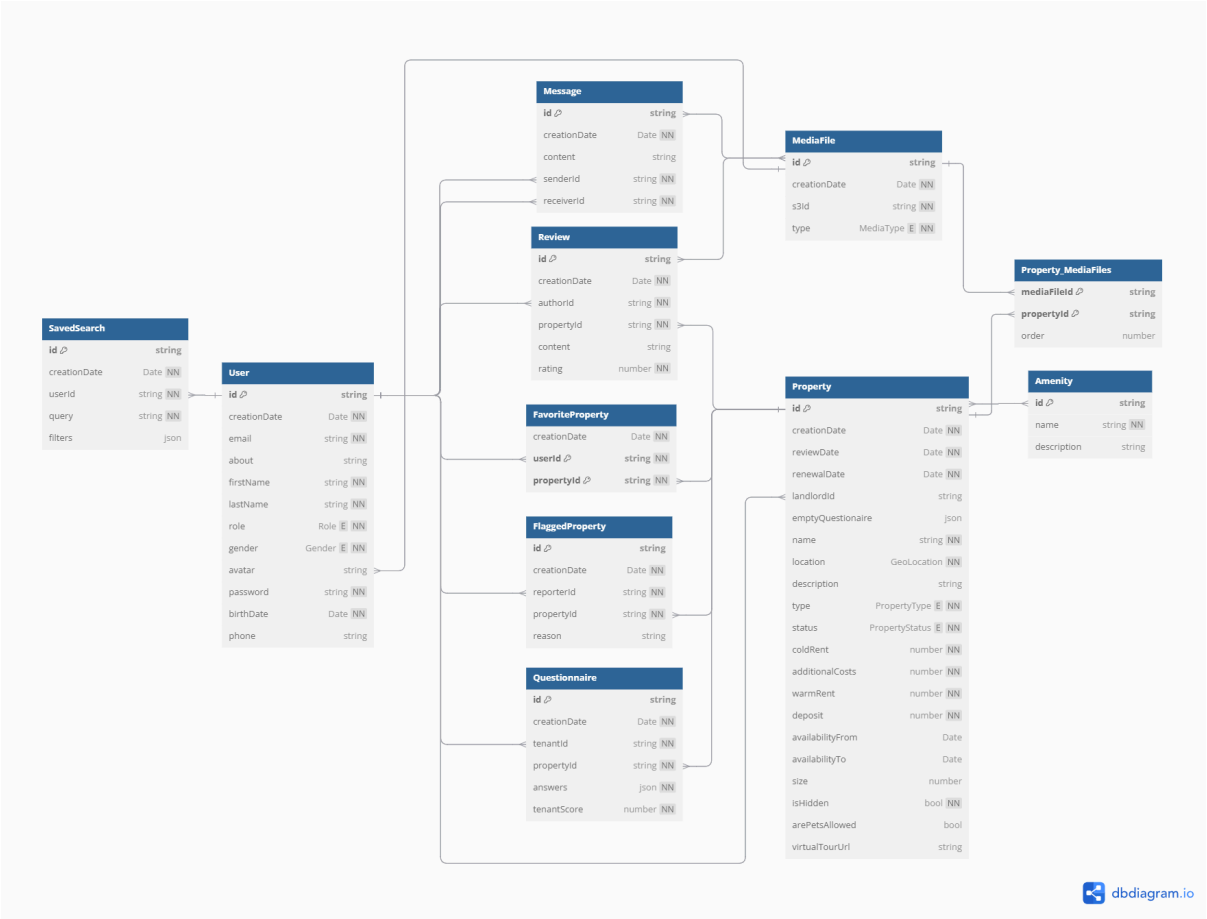





Moderator Dashboard
Welcome back, Julia Braun
Listings
Students
Landlords
My Profile
Listing Details
2-Bedroom Apartment near Campus
Review Pending
Approve
Reject
Contact
Description
A detailed description of the property
Address
Address of the property
Rent
Enter the total rent
Property Type
Type of Place - Studio Apt.
Availability Date
12-03-200 TO --
miro

11. High-Level Architecture & Database Organization

11.1 DB organization



11.2 Media storage

Azure blob storage will be utilized for media storage. It is similar to AWS S3 and consists of buckets (here they call it “containers”).

Storing files with binary data in a database does not provide any value for our project. Using an S3-like storage for files offers the ability to rapidly scale up and does not slow down our database.

11.3 Search/Filter Architecture and Implementation

We implement search/filter functionality using MySql for students to find the apartments easily. This feature will allow students to search/filter based on:

- Location: For example: searching for apartments in "Fulda"
- Price: Filter listings within a specific price range.
- Availability time: Check when an apartment is available.

- Type: Filter by shared or private apartments.
- Creation date: Sort or filter based on when the listing was created.
- Size: Look for apartments within a specific size range.

For text-based searches like property name or location, we will use the mySql **ILIKE** operator. This makes it possible to search without worrying about uppercase or lowercase letters.

To search an apartment based on the location “Fulda”

Query: SELECT * FROM property WHERE location ILIKE '%search_location%';

For numeric filters like warmrent and size, we will use the **BETWEEN** operator.

To find apartments priced between 300 and 400:

Query: SELECT * FROM property WHERE warmrent BETWEEN ? AND ?;

11.4 APIs

User APIs: -

- POST /api/users/register: Register a new user.
- POST /api/users/login: Authenticate a user.
- GET /api/users/profile: Retrieve user profile.
- PUT /api/users/profile: Update user profile.

Properties APIs: -

- GET /api/properties/{filters}: Retrieve a list of properties, with optional filters.
- GET /api/properties/:id: Fetch details of a specific listing.
- POST /api/properties: Add a new listing.
- PUT /api/properties/:id: Update an existing listing.
- DELETE /api/properties/:id: Delete an listing.

Search and Filter APIs: -

- GET /api/search: Perform a search query with parameters like location, price range, and etc...

Messages APIs: -

- GET /api/messages/:conversationId: To fetch all the interactions on a listing by two users.
- POST /api/messages: Submit a new message in a conversation.

Location APIs

- GET /api/locations/nearby: Retrieve nearby properties or landmarks based on user location.
- POST /api/locations/distance: Calculate the distance between two locations.

Reviews APIs: -

- POST /api/reviews/:propertyId: Submit a review for an accommodation.
- GET /api/reviews/:propertyId: Fetch reviews for a specific accommodation.

Notifications APIs: -

- POST /api/notifications/send: Send a notification to a user.
- GET /api/notifications/:userId: Retrieve notification history for a user.

Admin APIs: -

- GET /api/admin/users: Retrieve a list of all users.
- GET /api/admin/moderators: Fetch all moderators.
- GET /api/admin/properties: Retrieve all properties.
- DELETE /api/admin/:id: Delete any resource (users, bookings, properties) based on type.

Moderator APIs: -

- POST /api/moderator/users/approve: Approve a landlord.
- POST /api/moderator/users/deny: Deny a landlord.
- GET /api/moderator/unapproved-properties: Fetch all non-approved properties.
- GET /api/moderator/blacklisted: Retrieve all blacklisted properties.
- POST /api/moderator/messages/:propertyId: Retrieve the denial message for a property.

Flagged Properties APIs: -

- POST /api/flagged-properties/:propertyId: Flag a property for issues.
- GET /api/flagged-properties: Retrieve all flagged properties.

Saved Searches APIs: -

- GET /api/saved-searches/:userId: Retrieve saved searches for a user.

Questionnaire APIs: -

- POST /api/questionnaire/:propertyId: Submit a questionnaire for a property.
- GET /api/questionnaire/:propertyId: Retrieve questionnaire details for a property.

Analytics APIs: -

- GET /api/analytics/usage: Retrieve usage statistics (e.g., active users, popular listings).

11.5 Algorithms

We have decided to implement a basic and straightforward algorithm to calculate the tenant score for students, making it easier for landlords to prioritize tenants from a list. The landlord creates a questionnaire with specific questions and predefined answers, where each answer has a score associated with it. When a student fills out the questionnaire, each of their answers is compared to the predefined correct answers, and points are assigned accordingly. The overall tenant score is calculated by summing up the scores from each question, where each question's score is determined by the predefined answer chosen by the student. This approach ensures that landlords can quickly assess and compare potential tenants based on their responses.

Example:

Question	Answer options	Predefined Answer	Score	Student1	Student2
Stable Income	Yes/No	Yes	10	No	No
Previous Shared Housing Experienc e	Yes/No	Yes	8	Yes	No
Rental period	Less than a year / 1 year+	1 year+	10	Less than a year	1 year+

Student1 Score = (0 + 8 + 0) = 8

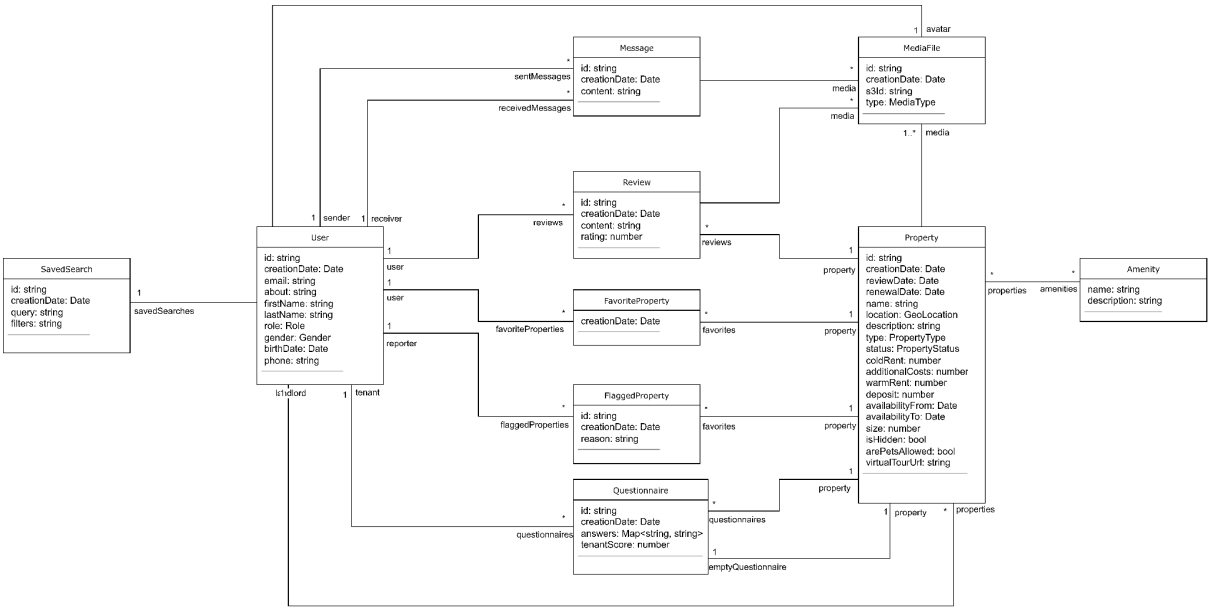
Student2 score = (0 + 0 + 10) = 10

11.6 Adjustments to software

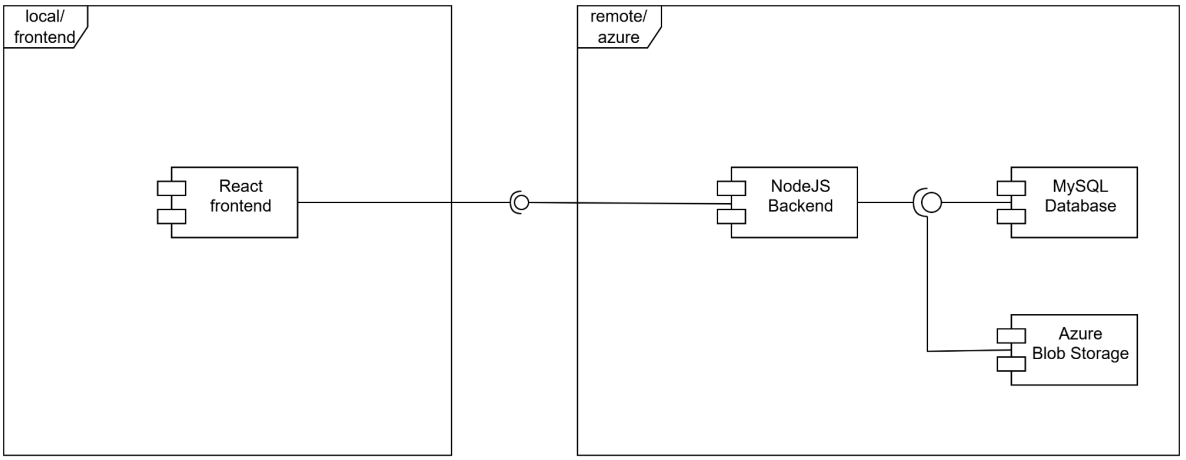
No adjustments have been made.

12. High-Level UML Diagrams

12.1 Class diagram



12.2 Deployment diagram



13. Key Risks

Skill Risks:

1. **Lack of Experience with Programming and Relevant Technologies**
    - **Risk**: One of our team members has no prior experience with JavaScript, React, Node.js, or programming concepts required for the project, which may delay the development process or hinder task completion.
    - **Mitigation:** Maira has proactively started watching tutorials and practicing relevant concepts to build foundational knowledge. Regular peer support and pair programming sessions can further aid in her learning curve.
2. **Uneven Skill Distribution**
    - **Risk:** Team members have varying levels of expertise in programming, which might lead to uneven workload distribution or delays in completing complex tasks.
    - **Mitigation:** Pair less experienced team members with more experienced ones for tasks. Allocate simpler tasks to beginners and ensure documentation and tutorials are available for self-paced learning.

Schedule Risks:

1. **Conflicting Schedules and Academic Workload**
    - **Risk**: Team members have varying class schedules, and major submissions or workloads in other subjects may conflict with the project timeline, potentially delaying progress.
    - **Mitigation**: Establish a shared project timeline and schedule weekly meetings to discuss progress and adjust tasks as needed. Prioritize clear communication and redistribute tasks during high-workload weeks to ensure consistent progress.
2. **Unexpected Absences from Meetings**
    - **Risk:** A team member may have an urgent personal or academic matter and be unable to attend a scheduled meeting, leading to missed updates or delays in decision-making.
    - **Mitigation:** Record important meetings and share notes to keep absent members informed. Assign a teammate to update them on key discussions and decisions. Use follow-up discussions or written updates to ensure they stay on track.

Technical Risks:

1. **Trouble with the cloud provider**
    - **Risk:** While unlikely, our cloud provider Azure could be down (e.g. for maintenance) while we need our instance to be running.
    - **Mitigation:** Check out other providers and prepare a backup plan.
2. **Integration Issues Between Frontend and Backend**
    - **Risk:** The frontend and backend teams might have different ideas about how the APIs should work, which could cause bugs or delays.
    - **Mitigation:** Have regular meetings to make sure both teams agree on the interfaces and keep clear documentation for the APIs. Start testing early and do it often to catch problems quickly.
3. **Learning Curve with New Tools or Frameworks**
    - **Risk:** Team members may take longer to become proficient with unfamiliar tools, frameworks, or libraries.
    - **Mitigation:** Identify required tools early, and encourage team members to take tutorials or participate in knowledge-sharing sessions.

Teamwork Risks:

1. **Coding style differences**
    - **Risk:** It is very common for members in a team to have different coding styles. Since oftentimes there is no “correct style”, this can lead to multiple styles in one project which can make the code hard to read.
    - **Mitigation:** Discuss a style to be used for this project and create a “prettier” or “eslint” config to be used for automatic formatting.
      E.g. using single/double quotes, end lines with semicolons, etc.
2. **Communication Tool Misalignment**
    - **Risk:** Team members may be used to different communication tools and miss important messages or notifications in the tool we’re using.
    - **Mitigation:** Agree on one primary communication tool (e.g., Discord) and remind everyone to check it regularly. Help team members set up notifications and provide quick guidance if needed.

Legal/Content Risks:

1. **Paid images & icons are used in the project**
    - **Risk:** Unfree images are used to illustrate certain properties and show their interior. While highly unlikely, a copyright holder could stumble upon the website to check for an apartment and report the infringement.
    - **Mitigation:** Check the licensing of all images before using them
2. **Transition from Azure Free Credits to Paid Usage**
    - **Risk:** The lack of clear communication about the switch from Azure's free credits to paid usage for a student account could lead to unexpected charges, which might create a legal risk.

- **Mitigation:** Stopping the server and contacting Azure support to resolve the billing issue can help prevent further unexpected charges.

14. Project Management

For Milestone 2, we divided our team into two groups: **Frontend** and **Backend**. This way, each group could focus on their own tasks while making sure we all stay connected on the main goals.

For communication, we use **Discord**, where we have a general channel for everyone to share updates and ask questions. We also have separate channels for **Frontend** and **Backend** teams to focus on their specific work. Each team holds its own meetings to discuss and assign tasks, ensuring that each task has a clear deadline. Afterward, we come together in joint meetings to review progress, address any challenges, and ensure everything is on track.

To manage tasks, we use **Trello**. We created a board with sections such as "To Do," "In Progress," "Done," and "Closed." Each task is broken down into smaller steps, assigned to team members, and given a deadline. This allows everyone to see who is working on what and track the overall progress of the project.

We hold regular meetings to check on the progress, talk about any challenges, and make sure everything is on track. These meetings help us stay organized and finish tasks on time.

Moving forward, we'll keep using **Trello** for task management and **Discord** for team communication. We will also have **weekly check-ins** to review our progress and make sure we are meeting deadlines.

By using these tools and staying in touch, we are confident that both teams will work efficiently and that the project will stay organized and on schedule.

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SSH URL	gdsd.norwayeast.cloudapp.azure.com
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